FACULTY OF ENGINEERING AND TECHNOLOGY

ASSIGNMENT SUBMISSION FORM

PROGRAMME: Bachelor of Information Systems (Honours) (Data Analytics)
SEMESTER: Apr 2026
COURSE: STAT2044 Statistics for Data Analytics
ASSESSMENT: Assignment Final Report
DEADLINE: 10 August 2026, 11.59 pm MYT

INSTRUCTIONS TO CANDIDATES

- This is a group submission as part of the requirements of the group project. Only one submission per group.

IMPORTANT
The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

Group Members

List down the name of the group members and the student IDs here.

Academic Honesty Acknowledgement

"We verify that this paper contains entirely our own work. We have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, we have not copied or inadvertently copied ideas, sentences, or paragraphs from another student(s). I realize the penalties *(refer to Student Handbook)* for any kind of copying or collaboration on any assignment."

Use of Generative AI Tools

If generative AI tools were used in the submitted work, acknowledge its use as follows:

This work was assisted by ChatGPT and Google Gemini for academic language improvement, guidance for data analysis and statistical interpretation.

…….
(Students' signature / Date)

Data Protection
The protection of personal data is an important concern to Sunway University and any personal data collected on this form will be treated in accordance with the Personal Data Protection Notice of the institution.
http://sunway.edu.my/pdpa/notice_english (English version)
http://sunway.edu.my/pdpa/notice_bm (Malay version)

Table of Contents

1. Introduction 3
1.1 General Description of the Dataset 3
1.2 Relationship with Sustainable Development Goals (SDGs) 4
1.3 Objectives of Data Analysis 5
1.4 Data Pre-Processing 5
2. Descriptive Analysis 6
3. Statistical Inference and Modelling 11
3.1 t-Test for Body Mass Index (BMI) and Family History of Overweight 11
3.2 One- way ANOVA Test for Body Mass Index (BMI) across different Physical Activity (Phy) Levels 14
3.3 Multiple Linear Regression Model 18
4. Conclusion 22
5. Analysis Output 23
References 49

1. Introduction

1.1 General Description of the Dataset

The dataset chosen for this study is the "Estimation of Obesity Levels Based on Eating Habits and Physical Condition" dataset from the UCI Machine Learning Repository, created by Palechor and de la Hoz Manotas (2019) to estimate the prevalence of obesity in people from Mexico, Peru, and Columbia depending on their eating patterns and physical state. Data were collected via a web-based survey, with additional synthetic data generated using the Weka tool and SMOTE technique to improve dataset balance (Palechor & de la Hoz Manotas, 2019).

The dataset comprises 2111 observations with 16 input variables and one target variable, 'NObeyesdad', which represents an individual's obesity classification. The variables can be classified into several groups. The demographic and physical characteristics, which are 'Gender', 'Age', 'Height' and 'Weight'. The eating habit-related variables, which are 'FAVC', 'FCVC', 'NCP', CAEC' and 'CALC'. Meanwhile, the lifestyle and behavioural factors include 'FAF', 'TUE', 'SMOKE', 'CH2O', 'SCC' and 'MTRANS'. Furthermore, the variable 'family_history_with_overweight' is accounted for to examine any potential genetic influences on the degree of obesity (Palechor & de la Hoz Manotas, 2019). The target variable, NObeyesdad, is a classification of individuals into seven obesity categories, namely insufficient weight, normal weight, overweight level I, overweight level II, obesity type I, obesity type II, and obesity type III (Palechor & de la Hoz Manotas, 2019).

Table 1: Summary and Description of Variables

Variable	Type	Description
Gender	Categorical	—
Age	Continuous	—
Height	Continuous	—
Weight	Continuous	—
family_history_with_overweight	Binary	Has a family member suffered or suffers from overweight?
FAVC	Binary	Do you eat high caloric food frequently?

FCVC	Integer	Do you usually eat vegetables in your meals?
NCP	Continuous	How many main meals do you have daily?
CAEC	Categorical	Do you eat any food between meals?
SMOKE	Binary	Do you smoke?
CH2O	Continuous	How much water do you drink daily?
SCC	Binary	Do you monitor the calories you eat daily?
FAF	Continuous	How often do you have physical activity?
TUE	Integer	How much time do you use technological devices such as cell phone, videogames, television, computer and others?
CALC	Categorical	How often do you drink alcohol?
MTRANS	Categorical	Which transportation do you usually use?
NObeyesdad	Categorical	Obesity level

Throughout the statistical analysis, the variable used as the primary target variable is continuous Body Mass Index (BMI), calculated as BMI = $\frac{Weight}{Height^2}$, instead of the derived Obesity Level categories (NObeyesdad) from the dataset. Analysing BMI directly avoids collapsing variance and unnecessary complexity from multinomial modelling, allowing linear models to retain the full granular spectrum while the continuous predicted values still directly corresponding to standard WHO obesity categories.

1.2 Relationship with Sustainable Development Goals (SDGs)

The dataset "Estimation of Obesity Levels Based on Eating Habits and Physical Condition" that analysed in this study closely aligned with the United Nations Sustainable Development Goal 3 (SDG 3): Good Health and Well-Being, which aims to ensure the healthy lives and promote well-being for the people in all ages (World Health Organization, 2024). Obesity is a major risk factors that will lead to non-communicable disease (NCDs), significantly increase the risk of cardiovascular disease, type 2 diabetes mellitus, hypertension and certain type of cancer (World Health Organization, 2024).

This study support SDG 3 by examining how lifestyle behaviours and demographic factors, such as physical activity, family history and other lifestyle-related factors to identify patterns that are associated with obesity. These findings can contribute to obesity prevention and help

reduce the burden of NCDs, thereby supporting improved public health outcomes (Hildebrand & Pfeifer, 2025).

1.3 Objectives of Data Analysis

Obesity is a complex health condition that is influenced by multiple demographics, behavioural and lifestyle factors. This study applies t-test, one-way Anova, multiple regression and data visualisation to examine obesity-related patterns and relationships within the dataset. The first objective is to examine the distribution of the obesity levels. The second objective aims to determine whether family history of overweight is associated with differences in mean BMI. Thirdly, this analysis determines whether the mean BMI differs across levels of physical activity frequency. Lastly, the analysis is used to identify which demographic, behavioural and lifestyle factors are most strongly associated with BMI through a multiple regression model. Ultimately, these insights aim to support obesity prevention strategies, healthier lifestyle choices and public health initiatives.

1.4 Data Pre-Processing

With the newly derived BMI variable used as the target variable, the data was cleaned and pre-processed to ensure data consistency. Age was rounded to the nearest integer, Height and Weight were rounded to one decimal point and BMI was rounded to the two decimal points. Ordinal attributes such as FCVC, NCP and CH2O contained unexpected six decimal points value, likely due to synthetic data generation, were rounded down to the nearest integer as they did not qualify for the next response category. Finally, labels were changed to more meaningful names to improve readability.

Table 2: Summary of Variable Renamed

Variable	Renamed Variable
BMI	BMI
Gender	Gender
Age	Age
Height	Height
Weight	Weight
family_history_with_overweight	FamHistory

FAVC	HighCalories
FCVC	Vege
NCP	MainMeal
CAEC	BetwFood
SMOKE	Smoke
CH2O	Water
SCC	MontCalories
FAF	Phy
TUE	Tech
CALC	Alcohol
MTRANS	Transport
NObeyesdad	ObeseLvl

2. Descriptive Analysis

Summary Statistics Results

The MEANS Procedure

Variable	Mean	Std Dev	Minimum	Maximum	Range	N	Lower Quartile	Median	Upper Quartile
BMI	29.7002	8.0113	12.9987	50.8118	37.8131	2111	24.3141	28.7191	36.0198
Age	24.3126	6.3460	14.0000	61.0000	47.0000	2111	19.9462	22.7779	26.0000
Height	1.7017	0.0933	1.4500	1.9800	0.5300	2111	1.6300	1.7005	1.7685
Weight	86.5861	26.1912	39.0000	173.0000	134.0000	2111	65.4239	83.0000	107.4827
Vege	1.2132	0.5994	0.0000	2.0000	2.0000	2111	1.0000	1.0000	2.0000
MainMeal	1.5234	0.8303	0.0000	3.0000	3.0000	2111	1.0000	2.0000	2.0000
Water	0.7125	0.5988	0.0000	2.0000	2.0000	2111	0.0000	1.0000	1.0000
Phy	0.7347	0.8328	0.0000	3.0000	3.0000	2111	0.0000	1.0000	1.0000
Tech	0.3813	0.5825	0.0000	2.0000	2.0000	2111	0.0000	0.0000	1.0000

Figure 1: Summary Statistic of Numerical Variables

Summary statistics includes all numerical variables across all 2111 observations. Figure 1 shows the summary statistics for numerical variables, including BMI, Age, Height, Weight, Vege, MainMeal, and Water, across all 2111 observations. The mean age of the participants was 24.32 years, and standard deviation (SD) was 6.35 years. The mean for Height was 1.70 m, with a SD of 0.09 m, while the mean for Weight was 86.59 kg, with a SD of 26.19 kg. The mean for BMI was 29.70 kg/m², and the SD was 8.01, showing relatively high variations in Weight and BMI. For lifestyle, the mean for Vege was 1.21, and the SD was 0.60 for the

lifestyle variables, MainMeal had a mean of 1.52 and a SD of 0.83, while Water had a mean of 0.71 and a SD of 0.60.

The SD indicates how much the observations vary from the mean. The higher the value of the SD, the more deviation there is, and conversely, the lower the value of the SD, the closer the value lies to the mean value. The largest SD was for weight (26.19) and the next largest was for BMI (8.01), suggesting more participant variation for these. The smallest SD was obtained for height, with participants' heights being relatively consistent.

In general, the value of the mean is fairly close to that of the median for most variables indicating that the distributions were not skewed to any great extent. There was, however, some differences between their values of mean and median in the case of Weight and BMI, which indicated some positive skewness.

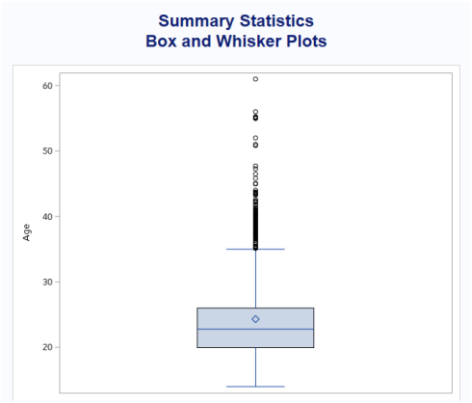


Figure 2: Boxplot for Age

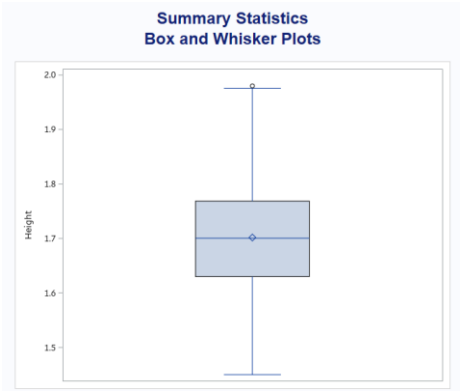


Figure 3: Boxplot for Height

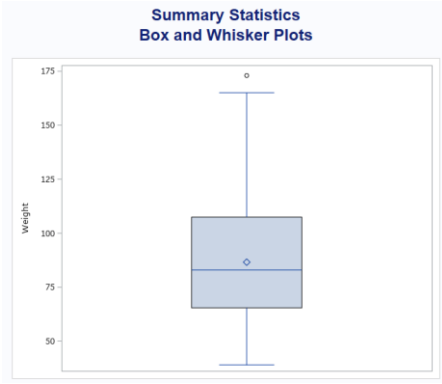


Figure 4: Boxplot of Weight

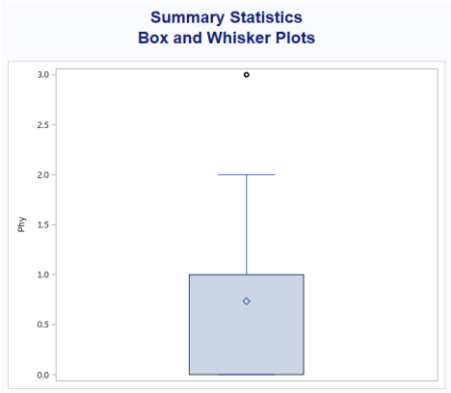


Figure 5: Boxplot for Phy

There were some extreme values (Figure 2 – Figure 4) which were observed but not removed from the dataset as they were considered to be real observations reflecting differences in the

physical size and lifestyle rather than data entry errors. Keeping these will help to maintain the natural variation and representativeness of the data set.

One-Way Frequencies
Results

The FREQ Procedure

Gender	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Female	1043	49.41	1043	49.41
Male	1068	50.59	2111	100.00

FamHistory	Frequency	Percent	Cumulative Frequency	Cumulative Percent
no	385	18.24	385	18.24
yes	1726	81.76	2111	100.00

HighCalories	Frequency	Percent	Cumulative Frequency	Cumulative Percent
no	245	11.61	245	11.61
yes	1866	88.39	2111	100.00

BetwFood	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Always	53	2.51	53	2.51
Frequently	242	11.46	295	13.97
Sometimes	1765	83.61	2060	97.58
no	51	2.42	2111	100.00

Smoke	Frequency	Percent	Cumulative Frequency	Cumulative Percent
no	2067	97.92	2067	97.92
yes	44	2.08	2111	100.00

Figure 6: One-way Frequencies Table for Gender, FamHistory, HighCalories, BetwFood and Smoke

MontCalories	Frequency	Percent	Cumulative Frequency	Cumulative Percent
no	2015	95.45	2015	95.45
yes	96	4.55	2111	100.00

Alcohol	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Always	1	0.05	1	0.05
Frequently	70	3.32	71	3.36
Sometimes	1401	66.37	1472	69.73
no	639	30.27	2111	100.00

Transport	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Automobile	457	21.65	457	21.65
Bike	7	0.33	464	21.98
Motorbike	11	0.52	475	22.50
Public_Transportation	1580	74.85	2055	97.35
Walking	56	2.65	2111	100.00

ObeseLvl	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Insufficient_Weight	272	12.88	272	12.88
Normal_Weight	287	13.60	559	26.48
Obesity_Type_I	351	16.63	910	43.11
Obesity_Type_II	297	14.07	1207	57.18
Obesity_Type_III	324	15.35	1531	72.52
Overweight_Level_I	290	13.74	1821	86.26
Overweight_Level_II	290	13.74	2111	100.00

Figure 7: One-way Frequencies Table for MontCalories, Alcohol, Transport and ObeseLvl

A one-way frequency analysis was used to examine the distribution of all categorical variables across all 2111 observations.

Most participants (1726, 81.76%) reported a family history of overweight, while 18.24% (385) reported that they did not. 1,866 of the participants (88.39%) said they ate HighCalories foods

while 245 participants (11.61%) did not. Most respondents (1765, 83.61%) reported to eat between meals “Sometimes”, while other respondents choose “Frequently” (242, 11.46%), “Always” (53, 2.51%), “No” (51, 2.42%).

For transportation option, public transportation was the most common mode of transport (1580, 74.85%). This is followed by automobile (457, 21.65%), walking (56, 2.65%), motorbike (11, 0.52%) and bicycle (7, 0.33%)).

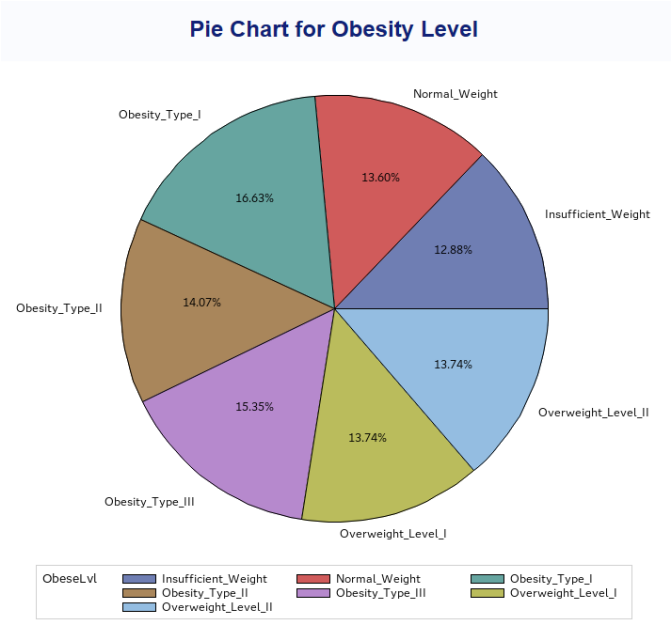


Figure 8: Pie Chart for Obesity Level

The variable ObeseLvl was relatively evenly distributed across it seven categories, as shown in the pie chart in Figure 8. The highest frequency was for Obesity Type I (351 participants or 16.63%) and the lowest frequency was for Insufficient Weight (272 participants or 12.88%), where no single category dominating the sample.

Overall, the frequency analysis of the data set reveals that there is a fairly even distribution across the ObeseLvl category, and that some categorical variables are heavily skewed in some categories, especially HighCalories, FamHistory and Transport.

The UNIVARIATE Procedure
Variable: BMI

Moments			
N	2111	Sum Weights	2111
Mean	29.7001588	Sum Observations	62697.0352
Std Deviation	8.01133661	Variance	64.1815143
Skewness	0.15416056	Kurtosis	-0.809085
Uncorrected SS	1997534.9	Corrected SS	135422.995
Coeff Variation	26.9740531	Std Error Mean	0.17436562

Basic Statistical Measures			
Location		Variability	
Mean	29.70016	Std Deviation	8.01134
Median	28.71909	Variance	64.18151
Mode	26.67276	Range	37.81307
		Interquartile Range	11.70564

Figure 9: Distribution Analysis of BMI

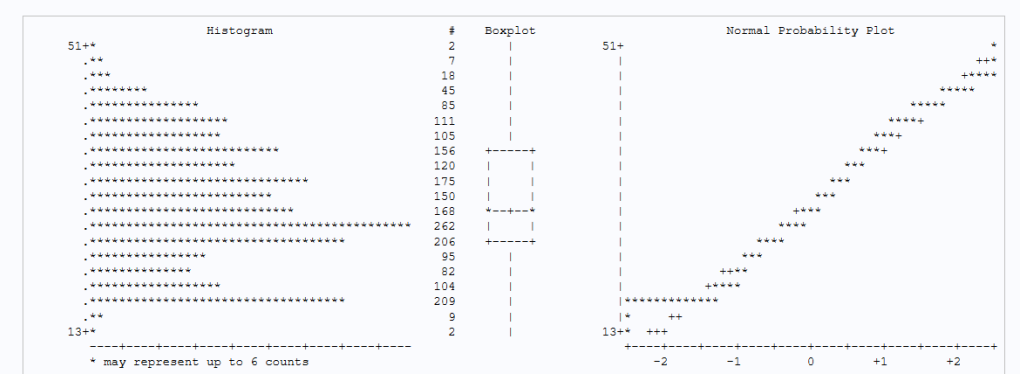


Figure 10: Histogram, Boxplot, and Normal Probability Plot for BMI

The distribution of BMI (Figure 9) was further examined using a normal probability plot, boxplot, histogram, skewness and kurtosis. The skewness of 0.154 indicates that BMI was slightly right skewed but lose to symmetric. This can also be validated by the small difference between mean and the median. Furthermore, the kurtosis of −0.809 indicates that the distribution is slightly flatter than normal.

The histogram (Figure 10) shows that the BMI values are concentrated in the middle with extremes. Although there are also some outlying values at both ends in the box plot, there were retained as genuine variation. The normal probability plot (Figure 10) demonstrates that BMI is approximately normally distributed with a slightly positive skew and the degree of departure from normality is not significant.

3. Statistical Inference and Modelling

3.1 t-Test for Body Mass Index (BMI) and Family History of Overweight

Hypothesis:

$$H_0: \mu_{\text{no family history}} = \mu_{\text{family history}}$$

(The mean BMI is the same between individuals with family history and without family history of overweight.)

$$H_1: \mu_{\text{no family history}} \neq \mu_{\text{family history}}$$

(The mean BMI is different between individuals with family history and without family history of overweight.)

Results and Interpretation:

**Note: M = Mean, SD = Standard Deviation, n = sample size*

Variable: BMI

FamHistory	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
no		385	21.5005	4.2107	0.2146	12.9987	35.0032
yes		1726	31.5292	7.4971	0.1805	13.2916	50.8118
Diff (1-2)	Pooled		-10.0287	7.0143	0.3953		
Diff (1-2)	Satterthwaite		-10.0287		0.2804		

FamHistory	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
no		21.5005	21.0786	21.9224	4.2107	3.9328	4.5311
yes		31.5292	31.1752	31.8831	7.4971	7.2551	7.7559
Diff (1-2)	Pooled	-10.0287	-10.8040	-9.2534	7.0143	6.8089	7.2326
Diff (1-2)	Satterthwaite	-10.0287	-10.5789	-9.4785			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	2109	-25.37	<.0001
Satterthwaite	Unequal	1007	-35.77	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	1725	384	3.17	<.0001

Figure 11: Welch Independent Samples t-Test Results for BMI by Family History of Overweight

To compare BMI between individuals with a family history of overweight and those without, an independent-samples Welch t-Test was performed. The assumption of equal variances was not met (Folded F = 3.17, $p < 0.0001$), meaning that the two groups had significantly different variability in their BMI values. Hence, the Welch t-Test was used.

Individuals with a family history of overweight had a significantly higher BMI (M = 31.53, SD = 7.50, n = 1726) than individuals without (M = 21.50, SD = 4.21, n = 385), t (1007) = -35.77, $p < 0.0001$. The average difference was -10.03 BMI units (95% CI: -10.58 to -9.48), which means that the BMI values in the family-history group were substantially higher.

The effect size was very large (Cohen's d ≈ 1.43), indicating that the difference is statistically significant and also of practical importance, with relatively little overlap between the two distributions of BMI. This is in line with the output histograms and box plots (Figures 12 and 13), which show a rightwards shift (higher BMI values) and a greater spread for the family-history group.

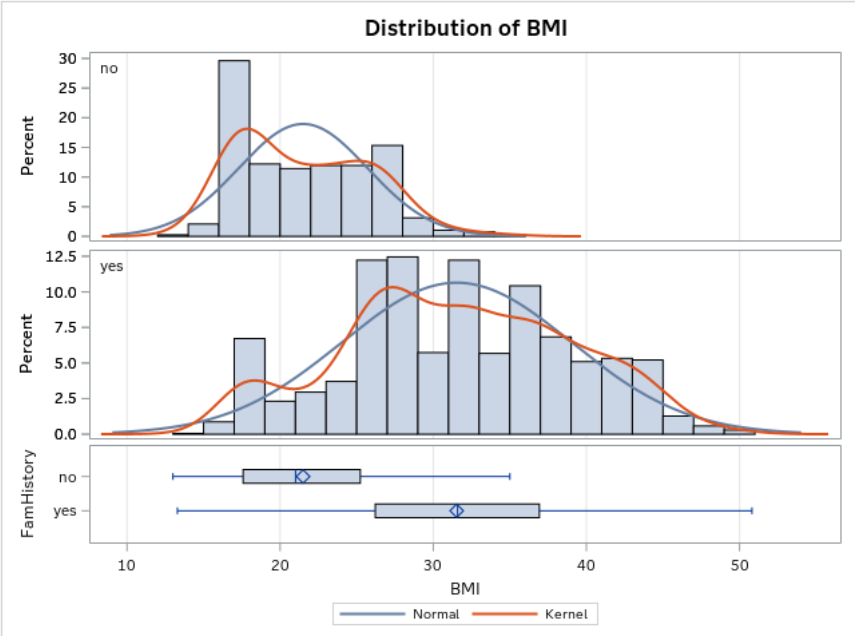


Figure 12: Distribution of Body Mass Index (BMI) for individuals with and without a family history of overweight

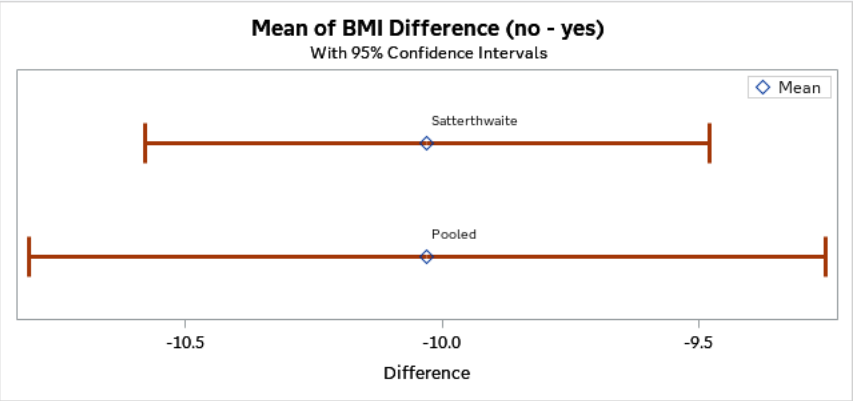


Figure 13: Comparison of BMI distributions between individuals with and without a family history of overweight using boxplots

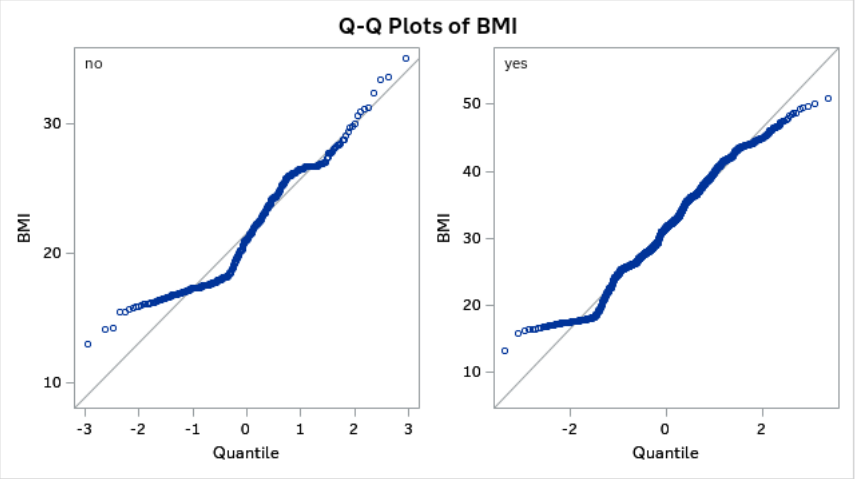


Figure 14: Assessment of normality for BMI in the family-history groups using normal Q–Q plots

The Q-Q plots (Figure 14) and histograms (Figure 12) show that there are some deviations from normality in both groups (right-skewness in the "no" group and a bimodal pattern in the "yes" group). However, the sample sizes are sufficiently large that the Welch t-Test is reasonably robust to these departures from the expected distribution.

Conclusion:

At the 5% significance level, we reject H_0. There is sufficient evidence to conclude that the mean BMI for individuals who have family history of overweight differs significantly from those who do not. Family history is strongly associated with BMI, with individuals who have a family history of overweight having a significantly higher BMI.

3.2 One- way ANOVA Test for Body Mass Index (BMI) across different Physical Activity (Phy) Levels

Hypothesis:

$$H_0 = \mu_{\text{Phy Level 0}} = \mu_{\text{Phy Level 1}} = \mu_{\text{Phy Level 2}} = \mu_{\text{Phy Level 3}}$$

(The mean BMI is the same across the different physical activity levels.)

H_1 = At least one mean BMI differs among the physical activity groups

(The mean BMI differs across the different physical activity levels.)

Results and Interpretation:

**Note: M = Mean, SD = Standard Deviation, n = sample size*

The GLM Procedure

Dependent Variable: BMI

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	15276.5865	5092.1955	89.30	<.0001
Error	2107	120146.4087	57.0225		
Corrected Total	2110	135422.9952			

R-Square	Coeff Var	Root MSE	BMI Mean
0.112806	25.42520	7.551324	29.70016

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Phy	3	15276.58655	5092.19552	89.30	<.0001

Parameter	Estimate		Standard Error	t Value	Pr > |t|
Intercept	24.43301580	B	0.87195184	28.02	<.0001
Phy 0	6.44619121	B	0.90371571	7.13	<.0001
Phy 1	6.61819457	B	0.91600238	7.23	<.0001
Phy 2	-0.63034340	B	0.97454786	-0.65	0.5178
Phy 3	0.00000000	B	.	.	.

Figure 15: The Effect of Physical Activity on BMI: One-way ANOVA Result

The one-way ANOVA was applied to test the BMI difference across different levels of physical activity (Phy). The output show that there is significant difference between means of BMI among the four physical activity groups of physical activity, $F(3, 2107) = 89.30$, $p < 0.0001$.

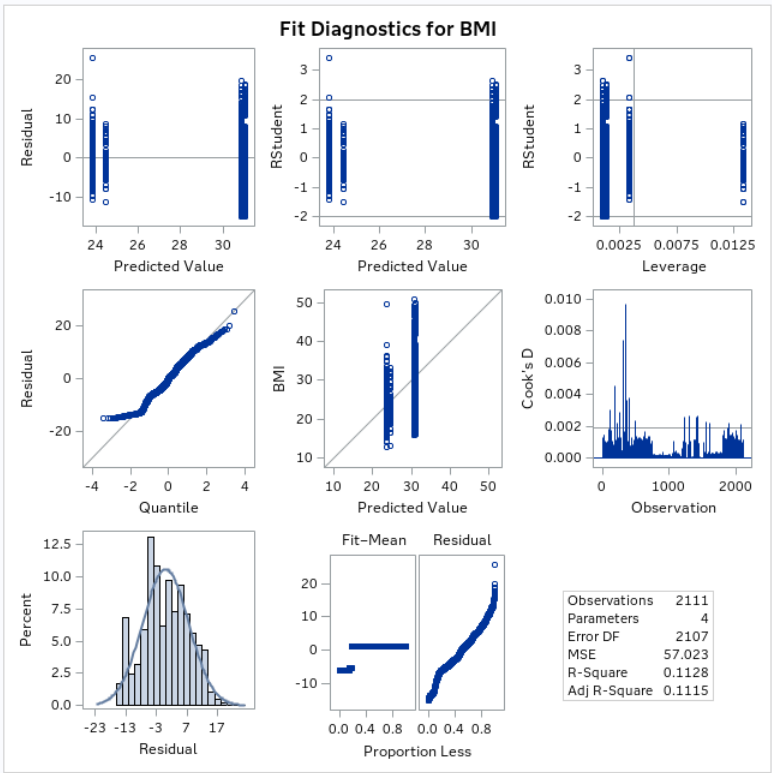


Figure 16: Fit Diagnostics for BMI in One-way ANOVA

The fit diagnostic plot (Figure 16) show that the residues deviate from normality in the tails and suggest possible heteroscedasticity. However, given the large sample size (n = 2111), the ANOVA remain robust to these violations and the findings remain reliable.

Linear Models

The GLM Procedure
Least Squares Means
Adjustment for Multiple Comparisons: Tukey-Kramer

Phy	BMI LSMEAN	LSMEAN Number
0	30.8792070	1
1	31.0512104	2
2	23.8026724	3
3	24.4330158	4

Least Squares Means for effect Phy
Pr > |t| for H0: LSMean(i)=LSMean(j)

Dependent Variable: BMI

i/j	1	2	3	4
1		0.9661	<.0001	<.0001
2	0.9661		<.0001	<.0001
3	<.0001	<.0001		0.9167
4	<.0001	<.0001	0.9167	

Figure 17: Least Square Means using Tukey-Kramer

Tukey-Kramer post hoc comparisons (Figure 17) revealed that Levels 0 and 1 were each significantly different from Levels 2 and 3 $p < 0.0001$. However, the differences between Levels 0 and 1 ($p = 0.9661$) and between Levels 2 and 3 ($p = 0.9167$) were not statistically significant. This suggests that very low and low activity levels yield similarly of high BMI, while moderate and high activity levels yield similarly lower BMI.

Linear Models

The GLM Procedure

Levene's Test for Homogeneity of BMI Variance
ANOVA of Squared Deviations from Group Means

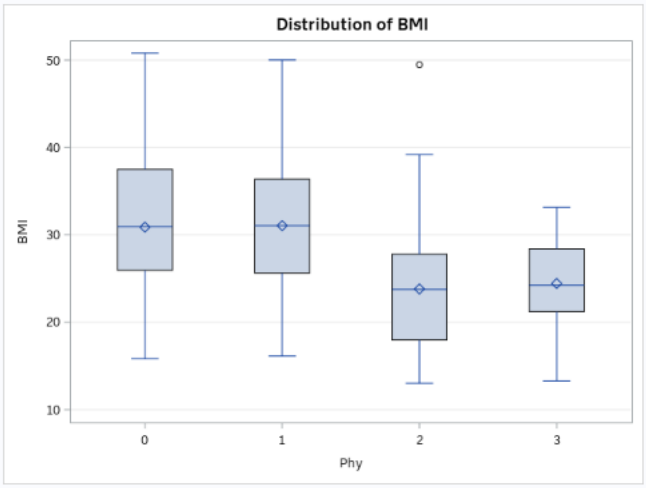
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Phy	3	367295	122432	29.61	<.0001
Error	2107	8710589	4134.1		

Figure 18: Levene's test

The Levene's test (Figure 18) showed the homogeneity of the variance was violated, $F(3, 2107) = 29.61$, $p < 0.0001$. However, given that the large sample size ($n = 2111$), the ANOVA remains robust to this violation.

Linear Models

The GLM Procedure



Level of Phy	N	BMI	
		Mean	Std Dev
0	1011	30.8792070	7.67377989
1	724	31.0512104	8.25111262
2	301	23.8026724	5.77157347
3	75	24.4330158	4.43399401

Figure 19: Figure 19: Distribution of BMI

The individuals in Physical Activity Levels 0 and 1 had higher mean BMI value (M = 30.88, SD = 7.67 and M = 31.05, SD = 8.25 respectively) than those individuals in Levels 2 and 3 (M = 23.80, SD = 5.77 and M = 24.43, SD = 4.43). The boxplot (Figure 19) also shows when the physical activity level increases, the mean BMI will generally decrease.

Conclusion:

At the 5% significance level, we reject the H_0. There is sufficient evidence to conclude that there is a difference in the mean value of BMI according to various levels of physical activity. Individuals who have low levels of physical activities have high BMI values compared to individuals with moderate to high levels of physical activities. Moreover, physical activity accounts for 11.3% variation in BMI (R^2 = 0.1128), implying that there may be other determinants of BMI. These findings have proved that there is a strong association between the level of physical activity and BMI.

3.3 Multiple Linear Regression Model

Overall Model Hypothesis

$$H_0: \beta_1 = \beta_2 = \beta_3 = \ldots = \beta_k = 0$$

(There is no linear relationship between the independent variables and BMI.)

$$H_1: \text{At least one } B_j \neq 0, j = 1, 2, 3, \ldots, k$$

(There is a linear relationship between at least one of the independent variables and BMI.)

**Note: M = Mean, SD = Standard Deviation, n = sample size*

An initial model, Model 1 (Appendix D), was fitted using all 17 variables. Referring to Table 3 in Appendix D, categorical variables that were transformed into ordinal or dummy variables and other numerical variables that are adjusted to have '0' as the baseline. The model was statistically significant with $F(17, 2093) = 7100.46$, $p < 0.0001$ and R^2 of 98.3%, but it was inflated due to the inclusion of Height and Weight, which are mechanically related to BMI.

Hence, Height, Weight and ObeseLvl were removed in Model 2 (Appendix E) to access lifestyle and behavioural factors independently. In this model, several remaining variables were found to be non-significant. The final regression model, Model 3 refines Model 2 by dropping FemaleGender ($p = 0.9765$), YesSmoke ($p = 0.6093$), and Water ($p = 0.3888$), all with $p > 0.05$. Then, the transportation variable is respecified. As transportation as no inherent order, it is split into two dummies according to the activeness involved. They are ActiveTrans (walking and bike) and PublicTrans (public transportation), with automobile and motorbike as the baseline (0 for both). This revealed that that public transport ($\beta = 4.062$) has a larger effect than active transport ($\beta = 1.840$), which was hidden in the combined dummy variable initially.

A machine generated stepwise regression that included ActiveTrans and PublicTrans (Appendix G) were generated to support Model 3. All other predictors continued to have the same sign, similar magnitude, and remained significant, which confirm that Model 3 is simpler and significant without changing the main conclusions.

Results and Interpretation:

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: BMI

Number of Observations Read	2111
Number of Observations Used	2111

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	12	67542	5628.46060	173.96	<.0001
Error	2098	67882	32.35539		
Corrected Total	2110	135423			

Root MSE	5.68818	R-Square	0.4987
Dependent Mean	29.70015	Adj R-Sq	0.4959
Coeff Var	19.15202		

Figure 20: Overall Linear Regression Results and Analysis of Variance (ANOVA) for predicting BMI

The final regression model was statistically significant with $F(12, 2098) = 173.96$ and $p < 0.0001$. The model can account for 49.87% of the variation in BMI ($R2 = 0.4987$), which is a good fit for this type of health data because the individual variability is relatively high. The adjusted R2 (0.4959) is very close to R2, implying that the model is not overfit. It is also confirmed that multicollinearity is not inflating parameter variance due to low Variance Inflation Factors (VIF) and condition index (Appendix F).

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	1	11.37518	1.08893	10.45	<.0001	9.23968	13.51067
Age	1	0.27298	0.02615	10.44	<.0001	0.22170	0.32425
YesFamHistory	1	7.39239	0.34391	21.50	<.0001	6.71795	8.06683
YesHighCalories	1	2.50848	0.41206	6.09	<.0001	1.70040	3.31656
Vege	1	3.21186	0.21603	14.87	<.0001	2.78820	3.63553
MainMeal	1	0.72543	0.15545	4.67	<.0001	0.42057	1.03028
Dummy_BetwFood	1	-3.61761	0.27756	-13.03	<.0001	-4.16193	-3.07330
YesMontCalories	1	-2.16256	0.61788	-3.50	0.0005	-3.37429	-0.95084
Phy	1	-1.14472	0.15825	-7.23	<.0001	-1.45507	-0.83438
Tech	1	-1.86346	0.22830	-8.16	<.0001	-2.31119	-1.41573
Dummy_Alcohol	1	1.52115	0.24687	6.16	<.0001	1.03702	2.00528
ActiveTrans	1	1.83983	0.86040	2.14	0.0326	0.15251	3.52716
PublicTrans	1	4.06182	0.38133	10.65	<.0001	3.31399	4.80964

Figure 21: Parameter estimates, standard errors, t-values, p-values, VIF and 95% confidence intervals for predictors of BMI

The fitted multiple linear regression equation predicting BMI is:

$$\begin{aligned}\widehat{BMI} = {} & 11.38 + 0.27(Age) + 7.39(YesFamHistory) + 2.51(YesHghCalories) \\ & + 3.21(Vege) + 0.73(MainMeal) - 3.62(Dummy_BetwFood) \\ & - 2.16(YesMontCalories) - 1.14(Phy) - 1.86(Tech) \\ & + 1.52(Dummy_Alcohol) + 1.84(ActiveTrans) + 4.06(PublicTrans)\end{aligned}$$

All 12 predictors were statistically significant at the 5% level. YesFamHistory had the strongest positive association with BMI at $\beta = 7.39$ and $p < 0.0001$, while Dummy_BetwFood had the strongest inverse relationship at $\beta = -3.62$ and $p < 0.0001$. Most predictors in the model aligned with expected behavioural patterns. Family history and high-calorie intake are primary drivers of higher BMI, while monitoring calorie intake and high frequency in physical activity contribute to lower predicted BMI values.

However, certain predictors yielded counterintuitive relationships with BMI. Higher vegetable consumption, active transportation and public transportation were unexpectedly associated with increased BMI, while higher usage of technology devices lowers the predicted BMI. These anomalous are likely due to unobserved factors and reverse causality. For example, individuals with higher BMIs practising healthier habits, unmeasured vegetable portion sizes, unknown walking distance and transit time, or unknown demographics impact.

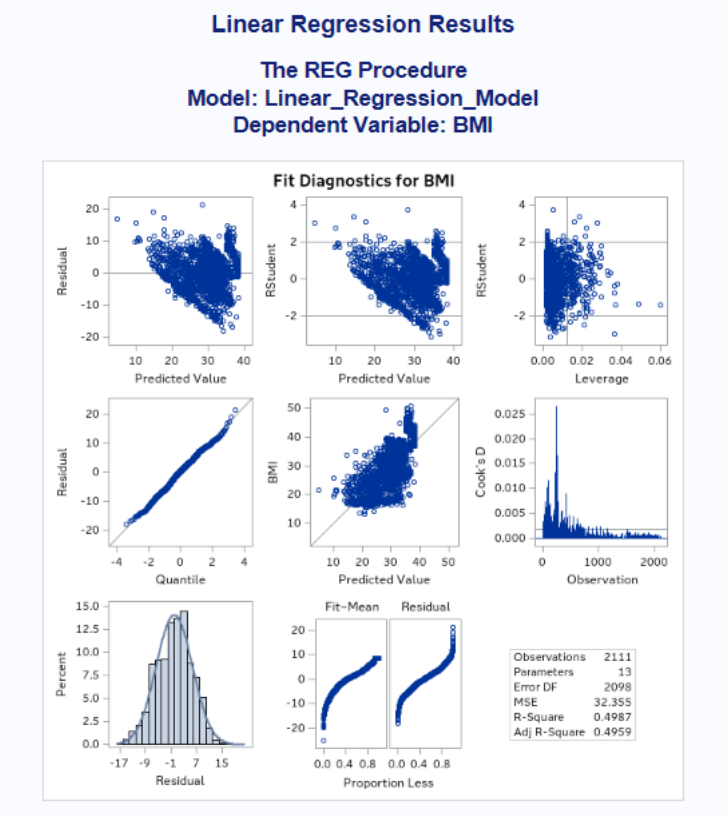


Figure 22: Diagnostic Plots for BMI Linear Regression Model

Statistical Modelling Problem:

The diagnostic plots (Figure 22) show residuals were reasonably close to the normal distribution in the central range with minor tail deviations. A downward pattern of the residual plot with unequal vertical thickness also suggesting that there may be heteroscedasticity. However, given the large sample size (n = 2111), the model remains robust for evaluating key predictors of BMI.

Conclusion:

Since $p < 0.0001$ is less than the significance level ($\alpha = 0.05$), we reject H_0. There is sufficient statistical evidence to support that at least one $Bj \neq 0$. There is at least one independent variable (Age, YesFamHistory, YesHighCalories, Vege, MainMeal, Dummy_BetwFood, YesMontCalories, Phy, Tech, Dummy_Alcohol, ActiveTrans, PublicTrans) that has a statistically significant linear relationship with BMI, together explaining 49.87% of variation in BMI.

4. Conclusion

To conclude, the demographic, behavioural and lifestyle factors associated with obesity were examined using the "Estimation of Obesity Levels Based on Eating Habits and Physical Condition" dataset (UCI Machine Learning Repository, 2019). The t-test shows that participants with a family history of overweight had a higher mean BMI compared to those without, suggesting that family history contributes significantly to obesity risk in this sample. The one-way ANOVA test showed the mean BMI across different physical activity frequency varies, with higher physical activity associated with lower mean BMI. This indicates that people that are more active in physical activity may have a lower risk of obesity. Multiple linear regression has identified that family history, eating patterns, vegetable consumption, physical activity, technology usage, alcohol consumption, and transportation mode significantly influence BMI.

However, several limitations were discovered throughout the analysis. As the dataset was partly synthetic data, the analysis should be interpreted as associations and not causation. Furthermore, it was observed that there was a possibility of heteroscedasticity in the regression residuals. Future improvements could include real-world and longitudinal data to validate these findings and strengthen evidence for obesity prevention strategies. Data transformations, such as $(Age)^2$, could be performed on variables to address linearity assumption violations.

Overall, these findings support SDG 3 goal by identifying modifiable lifestyle associated with obesity. Such outcomes could contribute to public health interventions in reducing obesity risk.

5. Analysis Output

The google drive link consists of the sources we used for analysis and the SAS EG document:

https://drive.google.com/drive/folders/1EcfUByTvG9FU_hQj7dunoqejKZfyCtZu?usp=sharing

SAS Enterprise Guide Process Flow

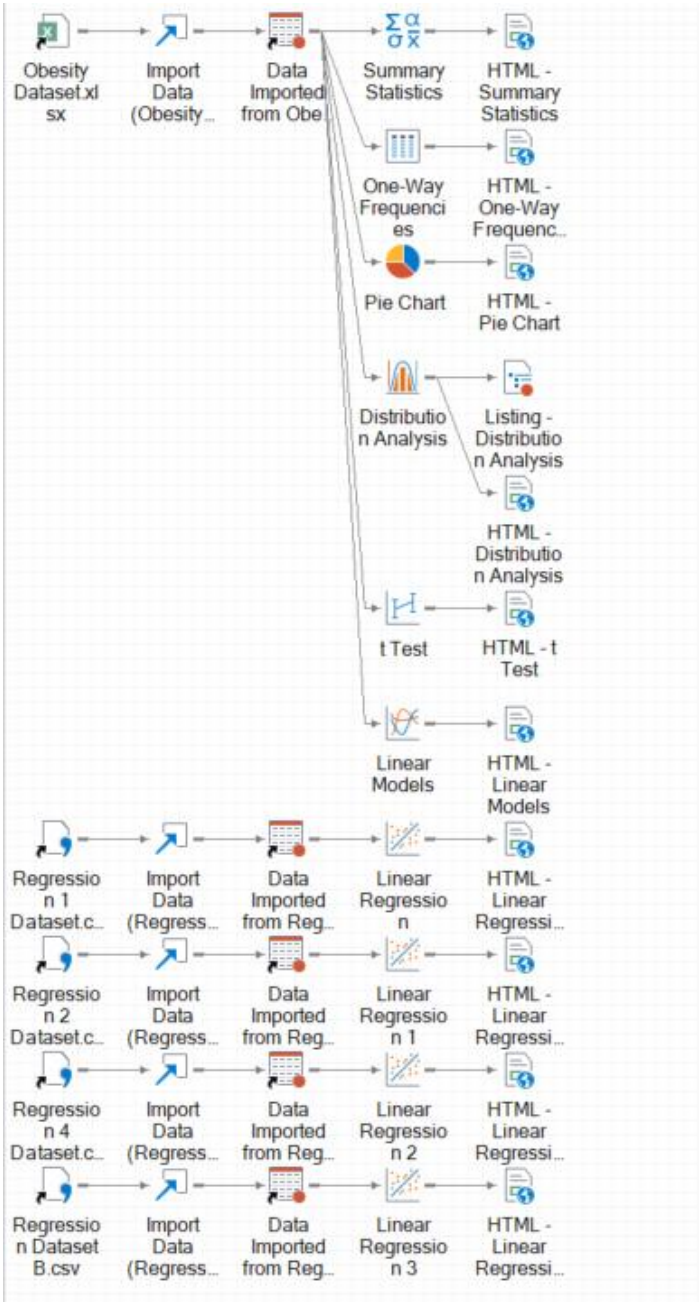


Figure 23: Process Flow

Appendix A

Descriptive Analysis Graphs

This appendix shows the histogram and box plots of the frequency of numerical variables of the dataset.

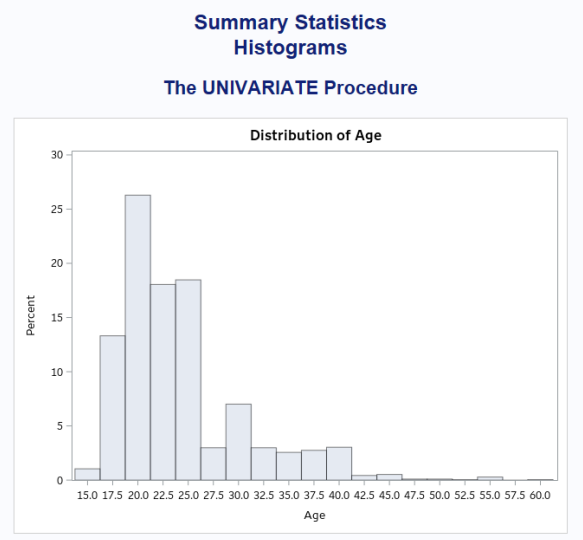


Figure 24: Histogram of Age

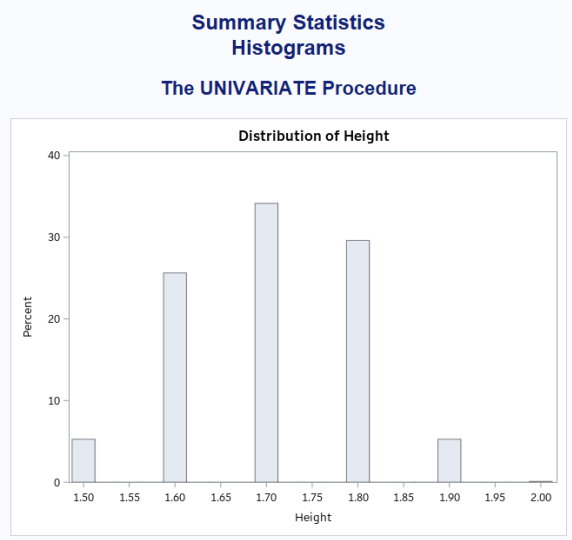


Figure 25: Histogram of Height

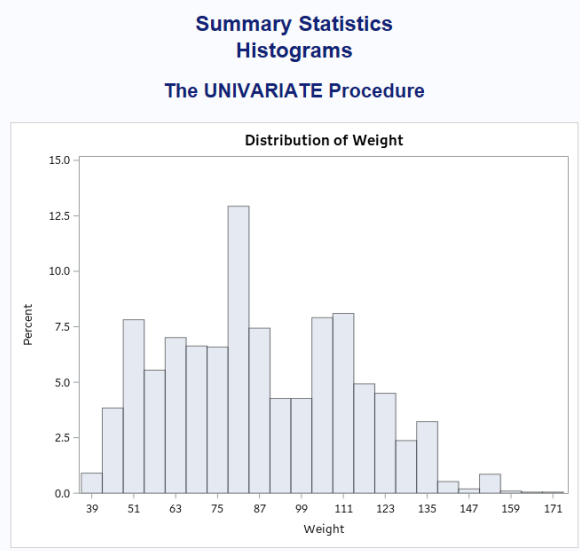


Figure 26: Histogram of Weight

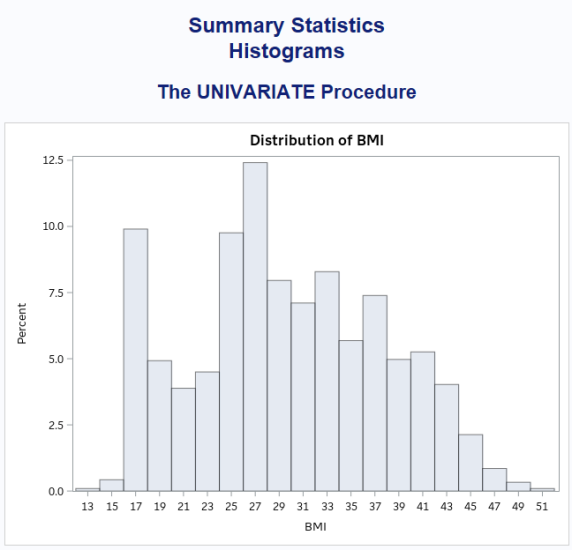


Figure 27: Histogram of BMI

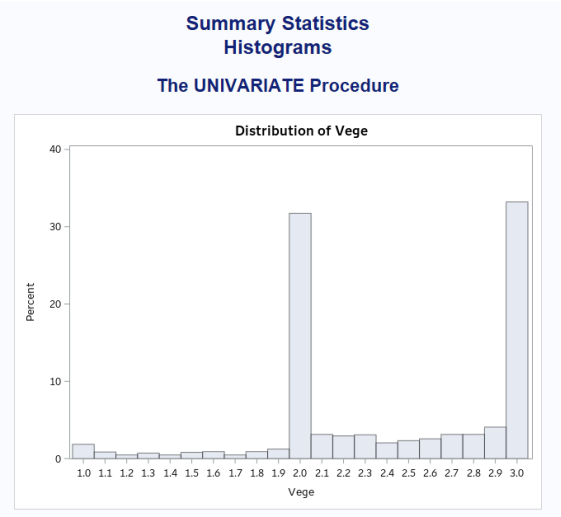


Figure 28: Histogram of Vege

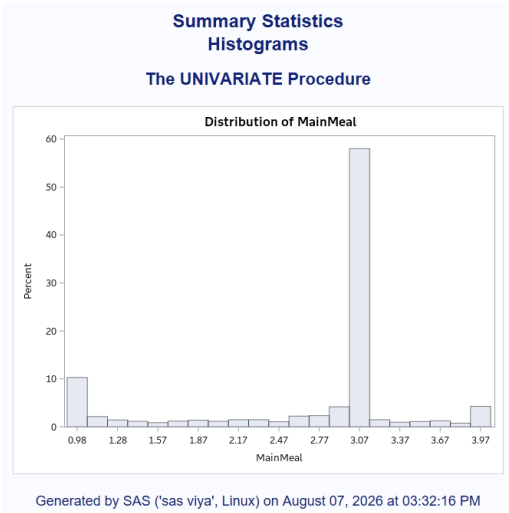


Figure 29: Histogram of MainMeal

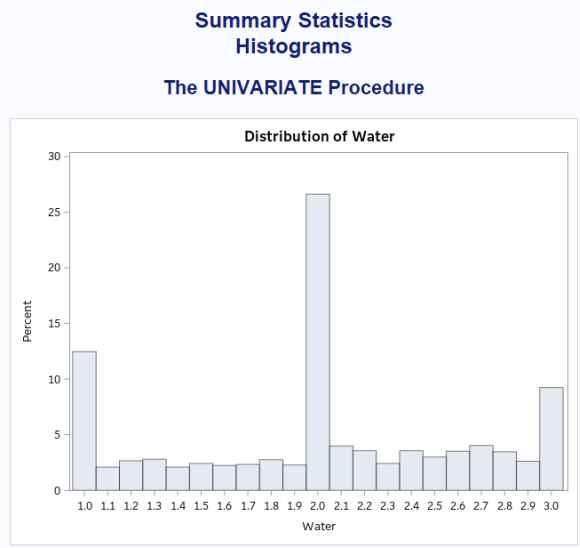


Figure 30: Histogram of Water

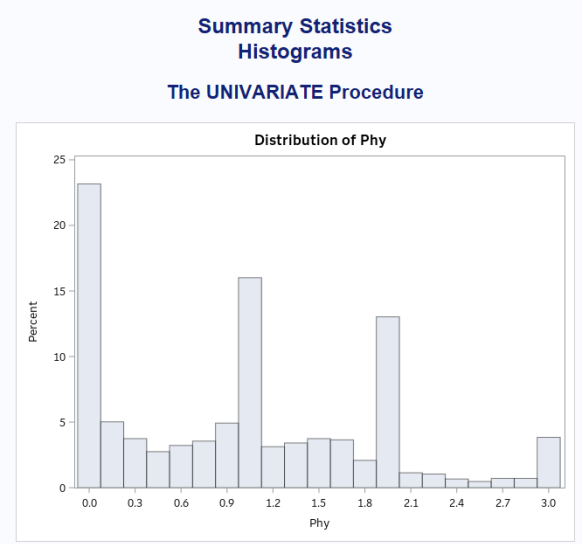


Figure 31: Histogram of Phy

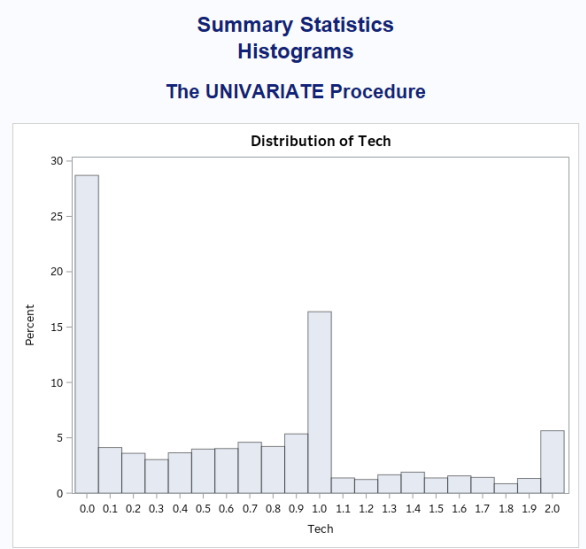


Figure 32: Histogram of Tech

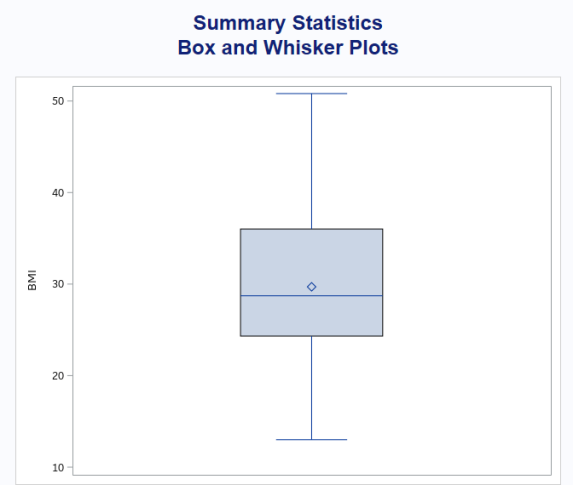


Figure 33: Boxplot of BMI

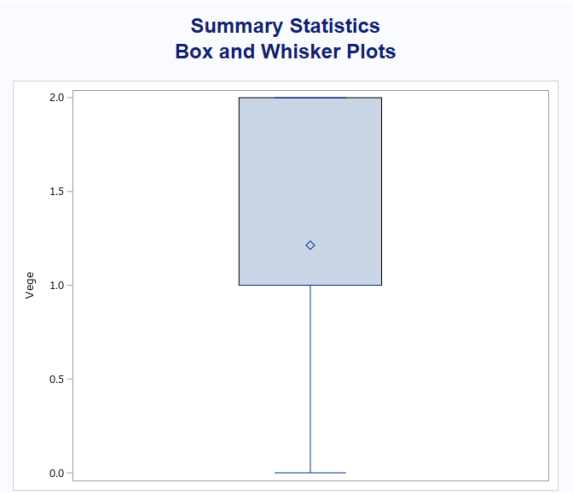


Figure 34: Boxplot for Vege

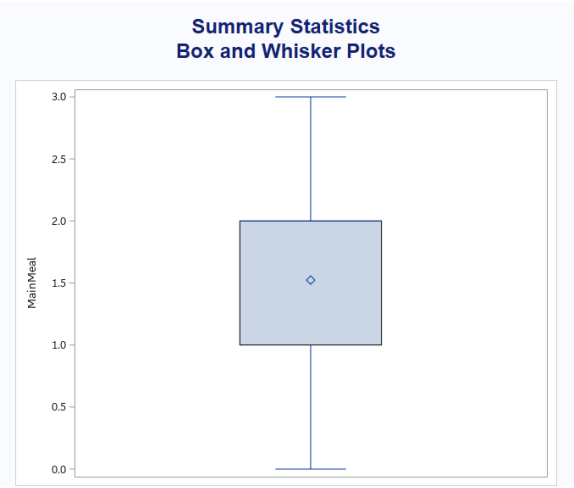


Figure 35: Boxplot for MainMeal

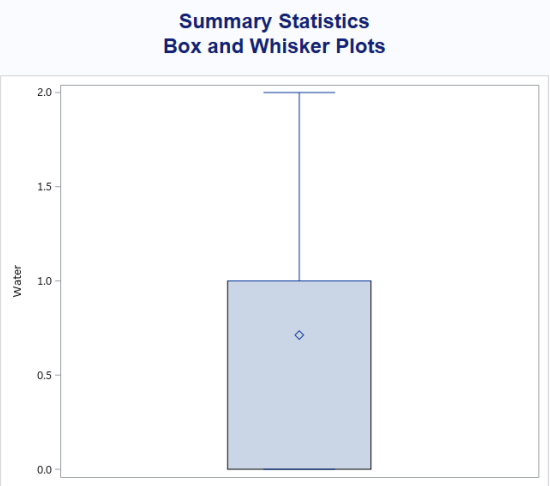


Figure 36: Boxplot for Water

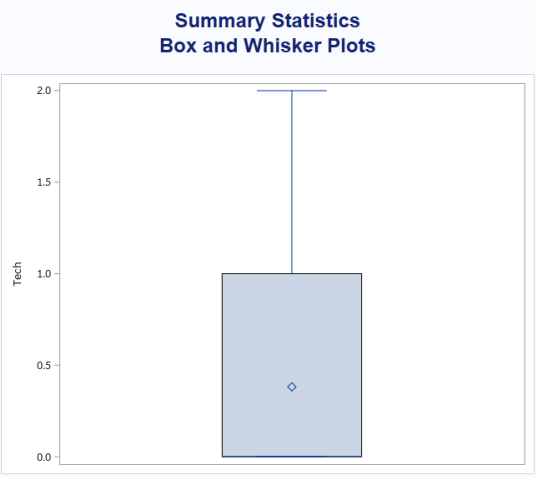


Figure 37: Boxplot for Tech

Appendix B

One-way Frequencies Table

This appendix shows the bar chart of the frequency of categorical variables of the dataset.

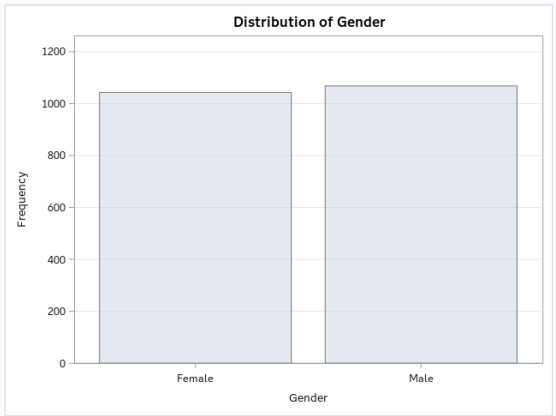


Figure 38: Bar Chart for Gender

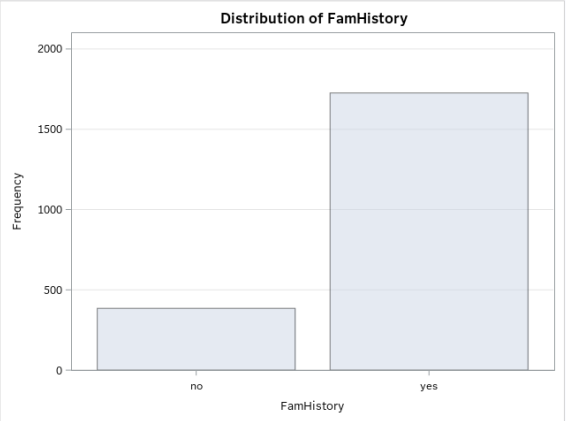


Figure 39: Bar Chart for FamHistory

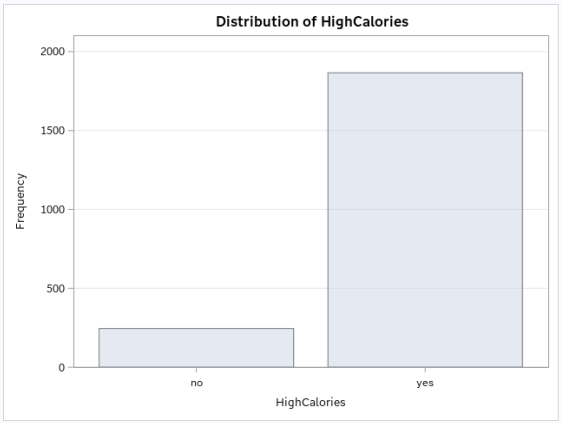


Figure 40: Bar Chart forHighCalories

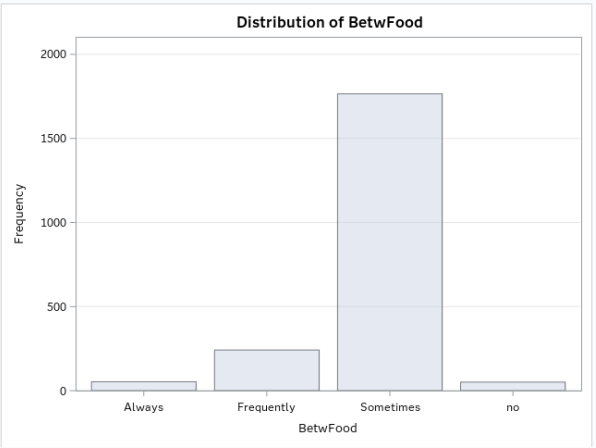


Figure 41: Bar Chart for BetwFood

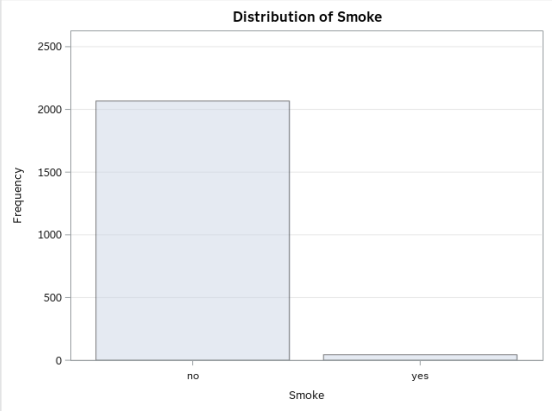


Figure 42: Bar Chart forSmoke

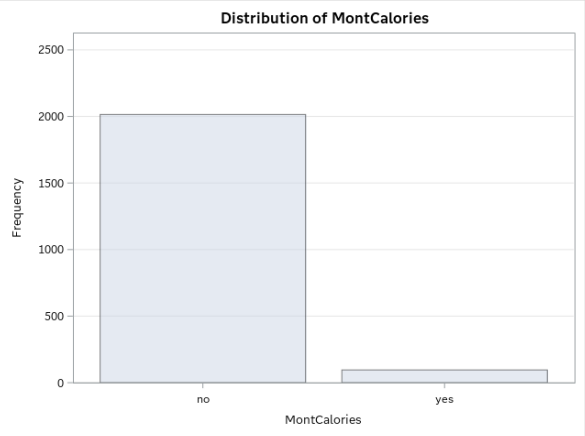


Figure 43: Bar Chart for MontCalories

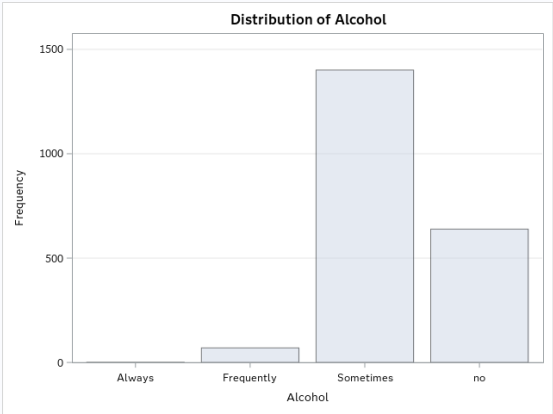


Figure 44: Bar Chart forAlcohol

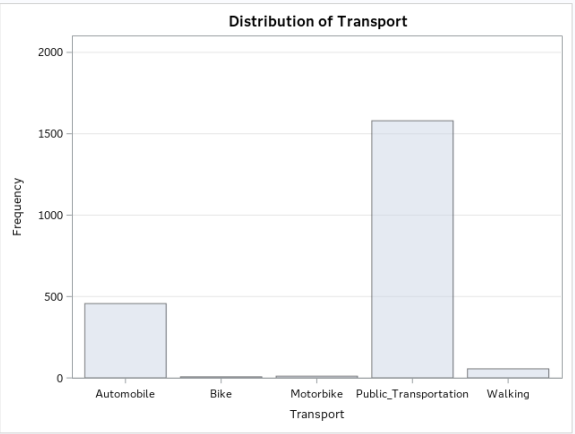


Figure 45: Bar Chart forTransport

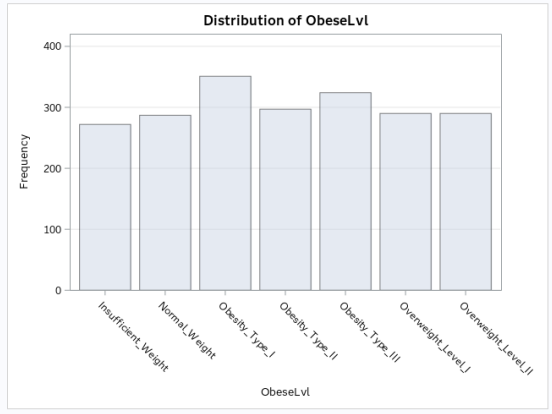


Figure 46: Bar Chart for ObeseLvl

Appendix C

One-way ANOVA

This appendix shows the class level information, LS-Means graph and BMI comparison graph for the variable, Phy.

The GLM Procedure

Class Level Information		
Class	Levels	Values
Phy	4	0 1 2 3

Number of Observations Read	2111
Number of Observations Used	2111

Figure 47: The Effect of Physical Activity on BMI: One-way ANOVA Result

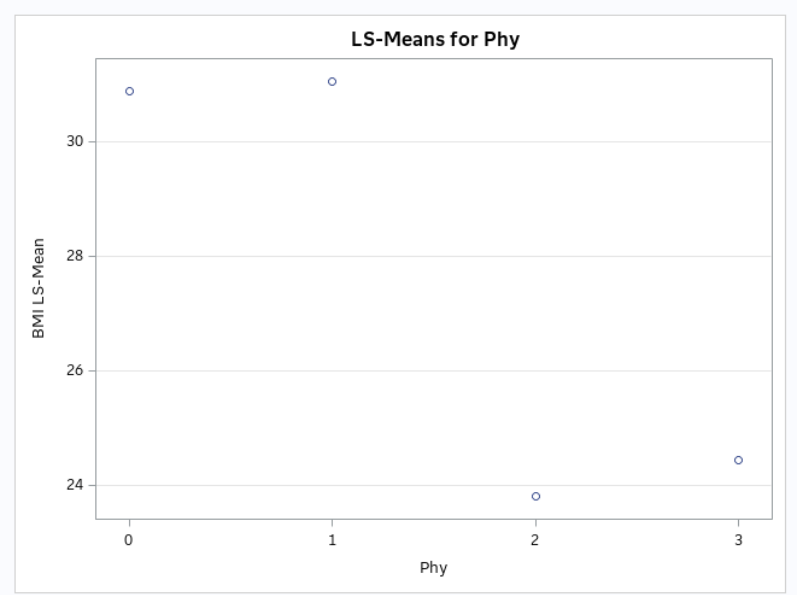


Figure 48: LS-Means for Phy

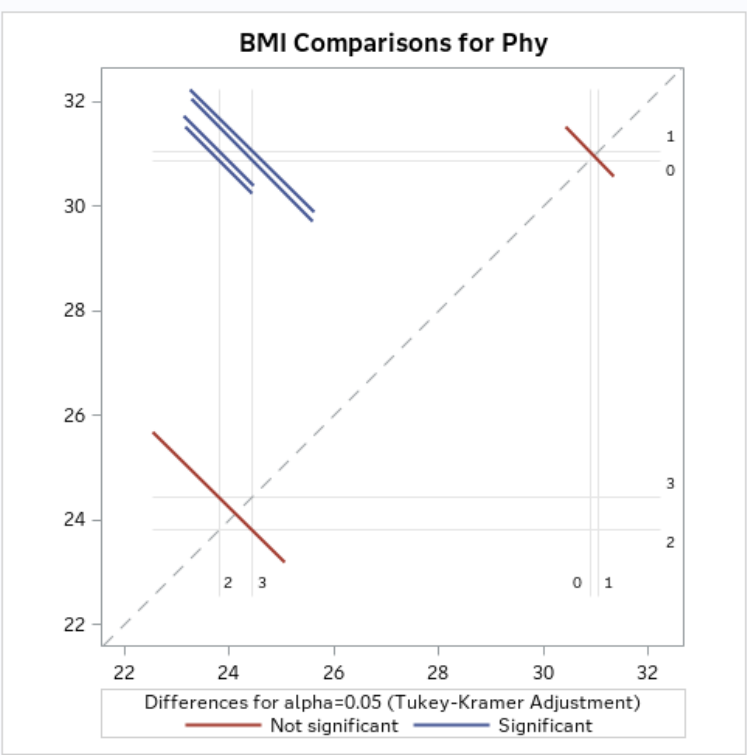


Figure 49: BMI Comparisons for Phy

Appendix D

Regression Model 1

This appendix shows the result output of Model 1, the initial model. It was developed by including all selected independent variables to examine their relationship with BMI. During the first regression, categorical variables were transformed into numerical formats before analysis. The High-Calorie Food Consumption, Vegetable Consumption and Between-Meal Eating Frequency were coded as ordinal variables, while the Transportation Method was converted into dummy variables with Automobile as the reference category.

Table 3: Summary of Dummy Variable Creation for Categorical Variables

Variables	Dummy Variables	Description
Gender	FemaleGender	If gender is female, shows '1', else shows '0'.
FamHistory	YesfamHistory	If participants have family history, shows '1', else shows '0'.
HighCalories	YesHighCalories	If participant eat high calories frequently, shows '1', else shows '0'.
BetwFood	Dummy_BetwFood	The number was assigned according to the frequency participants eat between meals like ordinal numerical variable, where no is '0', sometimes is '1', always is '2', always is '3'.
Smoke	YesSmoke	If participant smokes, shows '1', else shows '0'.
MontCalories	YesMontCalories	If participant monitor calories, shows '1', else shows '0'.
Alcohol	Dummy_Alcohol	The number was assigned according to the frequency participants consume alcohol like ordinal numerical variable, where no is '0', sometimes is '1', always is '2', always is '3'.
Transportation	Dummy_Trans (will be improved after Model 2)	The number was assigned according to the type of transportation where automobile is '0', motorbike is '1', bike is '2', public transportation is '3', walking is '4'

ObeseLvl	Dummy_ObeseLvl	The number was assigned according to the type of transportation where insufficient weight is '0', normal weight is '1', overweight level I and II are '2', Obesity Type I, II and III are '3'

The above variables take '0' as the baseline. Other numerical variables were standardised to have '0' as the baseline as well.

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: BMI

Number of Observations Read	2111
Number of Observations Used	2111

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	17	133115	7830.29383	7100.46	<.0001
Error	2093	2308.13322	1.10279		
Corrected Total	2110	135423			

Root MSE	1.05014	R-Square	0.9830
Dependent Mean	29.70015	Adj R-Sq	0.9828
Coeff Var	3.53580		

Figure 50: Linear Regression Results (Analysis of Variance) for Model 1

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	1	42.29085	0.65685	64.38	<.0001	41.00272	43.57899
FemaleGender	1	0.39688	0.06306	6.29	<.0001	0.27321	0.52055
Age	1	0.01193	0.00504	2.37	0.0179	0.00205	0.02180
Height	1	-23.31489	0.40849	-57.08	<.0001	-24.11598	-22.51379
Weight	1	0.27949	0.00259	107.76	<.0001	0.27441	0.28458
YesFamHistory	1	0.38161	0.07181	5.31	<.0001	0.24079	0.52244
YesHighCalories	1	-0.08493	0.07734	-1.10	0.2722	-0.23660	0.06673
Vege	1	0.37208	0.04520	8.23	<.0001	0.28344	0.46072
MainMeal	1	0.17928	0.02944	6.09	<.0001	0.12155	0.23702
Dummy_BetwFood	1	-0.24884	0.05358	-4.64	<.0001	-0.35392	-0.14377
YesSmoke	1	-0.57398	0.16330	-3.51	0.0004	-0.89423	-0.25372
Water	1	0.03949	0.04110	0.96	0.3367	-0.04111	0.12009
YesMontCalories	1	-0.23958	0.11535	-2.08	0.0379	-0.46580	-0.01336
Phy	1	-0.12782	0.03115	-4.10	<.0001	-0.18891	-0.06673
Tech	1	0.10368	0.04315	2.40	0.0164	0.01905	0.18831
Dummy_Alcohol	1	-0.05537	0.04712	-1.18	0.2401	-0.14778	0.03704
DummyTransport	1	-0.03755	0.02397	-1.57	0.1173	-0.08455	0.00945
DummyObeseLvl	1	0.77490	0.04048	19.14	<.0001	0.69552	0.85428

Figure 51: Linear Regression Result (Parameter Estimates) of Model 1

As a result, the overall regression model was statistically significant, as $F(17, 2093) = 7100.46$, while $p < 0.0001$, which indicating that at least one predictor was significantly associated with

the BMI. The model shown an R^2 = 0.9830 and an adjusted R^2 = 0.9828. This indicated that 98.30% of the variation in BMI was explained by the predictors. However, the high explanatory power may be influenced by the inclusion of Height and Weight variables as these variables are directly involved in BMI calculation and show a strong mathematical relationship with BMI. Therefore, Height, Weight and Obesity Level were removed in Model 2 (Appendix E) to better isolate the contribution of behavioural and lifestyle predictors of BMI.

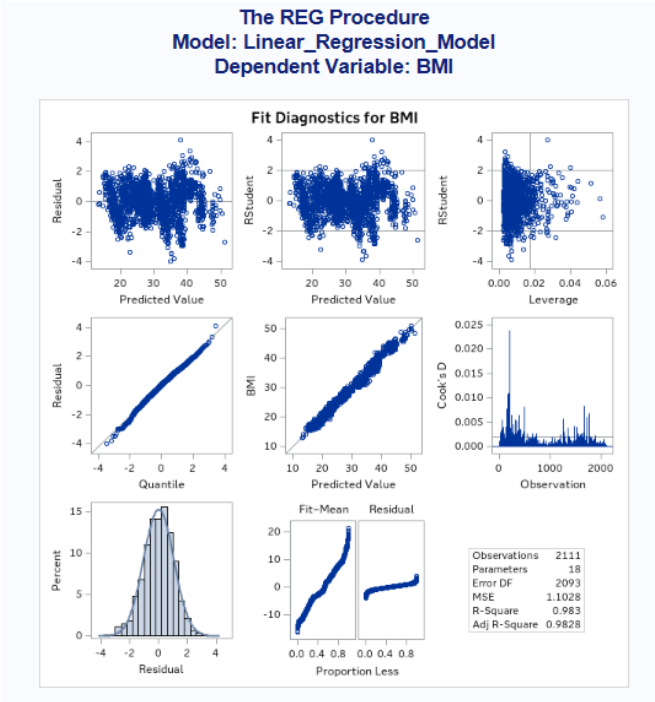


Figure 52: Linear Regression Result (Fit Diagnostics) for Model 1

Appendix E

Regression Model 2

This appendix shows the result output of Model 2. It was developed by removing the Height, Weight and Obesity Level variables from the initial model. Since BMI is calculated based on the Height and Weight, while obesity level represents weight status categories based on BMI. Therefore, including these variables as a predictor may create a direct mathematical relationship with the dependent variable.

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: BMI

Number of Observations Read	2111
Number of Observations Used	2111

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	14	67068	4790.57717	146.90	<.0001
Error	2096	68355	32.61214		
Corrected Total	2110	135423			

Root MSE	5.71070	R-Square	0.4952
Dependent Mean	29.70015	Adj R-Sq	0.4919
Coeff Var	19.22786		

Figure 53: Linear Regression Results (Analysis of Variance) for Model 1

After removing them, the overall of model 2 remained statistically significant, F (14, 2096) = 146.90, $p<0.0001$. The model produced an $R^2 = 0.4952$ and adjusted $R^2 = 0.4919$, indicating that the remaining predictors explained approximately 49.52% of the variation in BMI. However, three predictors, Female Gender ($p = 0.9765$), YesSmoke ($p = 0.6093$) and Water ($p = 0.3888$) were not significant at the 5% significant level. These variables were therefore removed in Model 3 to produce a more refined, concise and fully significant model.

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept	1	11.59416	1.11071	10.44	<.0001	9.41595	13.77238
FemaleGender	1	-0.00817	0.27768	-0.03	0.9765	-0.55272	0.53638
Age	1	0.26306	0.02629	10.01	<.0001	0.21151	0.31461
YesFamHistory	1	7.38838	0.34873	21.19	<.0001	6.70449	8.07227
YesHighCalories	1	2.80123	0.41189	6.80	<.0001	1.99348	3.60897
Vege	1	3.16188	0.22825	13.85	<.0001	2.71427	3.60949
MainMeal	1	0.70630	0.15614	4.52	<.0001	0.40009	1.01250
Dummy_BetwFood	1	-3.62831	0.28024	-12.95	<.0001	-4.17790	-3.07872
YesSmoke	1	-0.45367	0.88762	-0.51	0.6093	-2.19438	1.28705
Water	1	0.19229	0.22306	0.86	0.3888	-0.24516	0.62974
YesMontCalories	1	-2.21198	0.62418	-3.54	0.0004	-3.43605	-0.98791
Phy	1	-1.28943	0.16435	-7.85	<.0001	-1.61173	-0.96713
Tech	1	-1.99263	0.22877	-8.71	<.0001	-2.44126	-1.54399
Dummy_Alcohol	1	1.51826	0.24953	6.08	<.0001	1.02890	2.00761
DummyTransport	1	1.26688	0.12669	10.00	<.0001	1.01843	1.51532

Figure 54: Linear Regression Result (Parameter Estimates) for Model 2

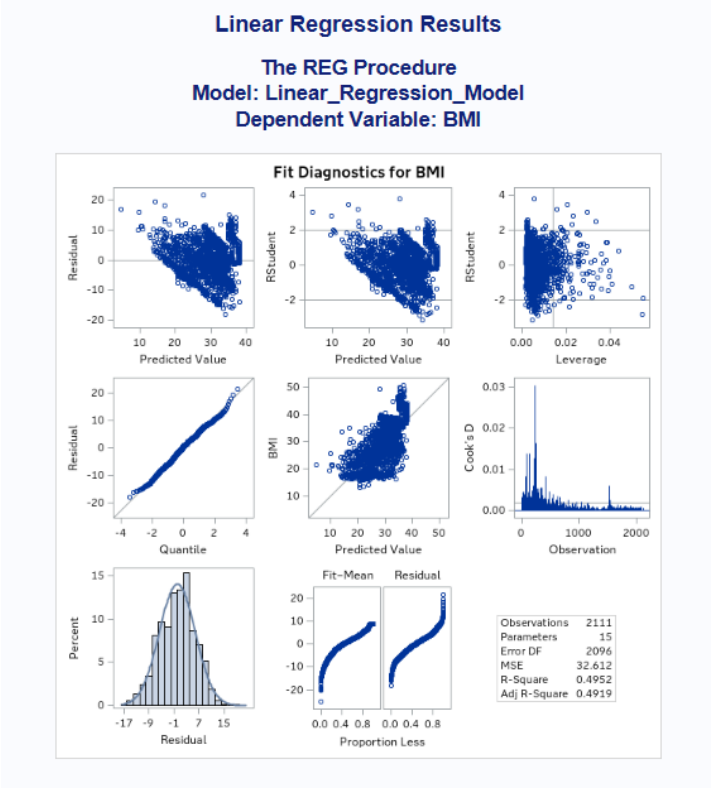


Figure 55:Linear Regression Result (Fit Diagnostics) for Model 2

Appendix F

Regression Model 3

This appendix shows the parameter analysis with Variance Inflation Factors (VIF) and collinearity table for Model 3.

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation	95% Confidence Limits	
Intercept	1	11.37518	1.08893	10.45	<.0001	.	0	9.23968	13.51067
Age	1	0.27298	0.02615	10.44	<.0001	0.55707	1.79509	0.22170	0.32425
YesFamHistory	1	7.39239	0.34391	21.50	<.0001	0.86905	1.15068	6.71795	8.06683
YesHighCalories	1	2.50848	0.41206	6.09	<.0001	0.87992	1.13647	1.70040	3.31656
Vege	1	3.21186	0.21603	14.87	<.0001	0.91451	1.09349	2.78820	3.63553
MainMeal	1	0.72543	0.15545	4.67	<.0001	0.92048	1.08639	0.42057	1.03028
Dummy_BetwFood	1	-3.61761	0.27756	-13.03	<.0001	0.90670	1.10291	-4.16193	-3.07330
YesMontCalories	1	-2.16256	0.61788	-3.50	0.0005	0.92486	1.08124	-3.37429	-0.95084
Phy	1	-1.14472	0.15825	-7.23	<.0001	0.88282	1.13273	-1.45507	-0.83438
Tech	1	-1.86346	0.22830	-8.16	<.0001	0.86694	1.15348	-2.31119	-1.41573
Dummy_Alcohol	1	1.52115	0.24687	6.16	<.0001	0.94684	1.05614	1.03702	2.00528
ActiveTrans	1	1.83983	0.86040	2.14	0.0326	0.80174	1.24728	0.15251	3.52716
PublicTrans	1	4.06182	0.38133	10.65	<.0001	0.55987	1.78614	3.31399	4.80964

Collinearity Diagnostics

Number	Eigenvalue	Condition Index	Proportion of Variation											
			Intercept	Age	YesFamHistory	YesHighCalories	Vege	MainMeal	Dummy_BetwFood	YesMontCalories	Phy	Tech	Dummy_Alcohol	ActiveTrans	PublicTrans
1	8.49175	1.00000	0.00017499	0.00044473	0.00181	0.00124	0.00207	0.00239	0.00158	0.00067864	0.00337	0.00291	0.00308	0.00036671	0.00148
2	1.12299	2.74986	0.00001885	0.00019718	0.00229	0.00170	0.00008456	0.00004994	0.00009063	0.14229	0.02724	0.04304	0.00216	0.41671	0.00326
3	0.93794	3.00892	5.096833E-7	0.00003114	0.00106	0.00019439	0.00010170	0.00013249	0.00003712	0.72895	0.00010233	0.01074	0.00004980	0.15824	0.00122
4	0.67928	3.53568	0.00002626	0.00080998	0.00217	0.00021251	0.00826	0.00087674	0.00009051	0.00670	0.01450	0.65552	0.01653	0.14747	0.00310
5	0.51106	4.07627	0.00002073	0.00015311	0.00192	0.00133	0.00004197	0.00137	0.00007057	0.02510	0.79192	0.09536	0.01841	0.05335	0.00145
6	0.29728	5.34464	0.00037189	0.00183	0.05593	0.00332	0.00999	0.00062635	0.00758	0.00605	0.02548	0.01313	0.84200	0.00492	0.00212
7	0.25371	5.78534	0.00001920	0.01533	0.05604	0.00631	0.03752	0.01680	0.00097355	0.02272	0.00004028	0.03507	0.00314	0.06923	0.36819
8	0.20791	6.39091	0.00045069	0.00242	0.08133	0.03758	0.05247	0.62155	0.02844	0.01274	0.06272	0.00028552	0.02556	0.01947	0.01670
9	0.17222	7.02197	0.00016676	0.00844	0.11332	0.00530	0.29776	0.30834	0.13454	0.02837	0.00055574	0.01189	0.00049733	0.04349	0.07427
10	0.14630	7.61859	0.00119	0.00346	0.10173	0.01686	0.52548	6.448002E-7	0.34908	0.00205	0.00600	0.09679	0.00367	0.01042	0.00596
11	0.10485	8.99950	0.00071164	0.00220	0.43738	0.50328	0.01551	0.01605	0.17449	0.00318	0.00020043	0.00075982	0.06132	0.00596	0.00400
12	0.06479	11.44857	0.01833	0.25797	0.14030	0.32654	0.04145	0.00298	0.24044	0.00382	0.00251	0.00318	0.02081	0.00234	0.07090
13	0.00993	29.24475	0.97852	0.70671	0.00473	0.09614	0.00926	0.02883	0.06258	0.01734	0.06535	0.03133	0.00276	0.06803	0.44735

Figure 56: Parameter Analysis and Collinearity Table for Model 3

Appendix G

Stepwise Regression Model

This appendix shows the result output of the stepwise multiple linear regression. It was performed as a validation analysis for Model 3. An automated procedure was used to identify a set of predictors that would yield similar results as in Model 3. As can be seen in Step 10, 12 predictors were found to be significant with no other variables meeting the entry criterion ($p < 0.15$).

Stepwise Selection: Step 1

Variable YesFamHistory Entered: R-Square = 0.2338 and C(p) = 1098.032

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	31660	31660	643.48	<.0001
Error	2109	103764	49.20035		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	21.50044	0.35748	177974	3617.32	<.0001
YesFamHistory	10.02873	0.39535	31660	643.48	<.0001

Bounds on condition number: 1, 1

Figure 57: Stepwise Regression Step 1

Stepwise Selection: Step 2

Variable Tech Entered: R-Square = 0.2999 and C(p) = 823.5270

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	40611	20306	451.47	<.0001
Error	2108	94812	44.97705		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	23.28141	0.36436	183630	4082.74	<.0001
YesFamHistory	9.50747	0.37980	28185	626.65	<.0001
Tech	-3.55270	0.25182	8951.91689	199.03	<.0001

Bounds on condition number: 1.0096, 4.0382

Figure 58: Stepwise Regression Step 2

Stepwise Selection: Step 3

Variable Vege Entered: R-Square = 0.3487 and C(p) = 621.2499

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	47225	15742	376.06	<.0001
Error	2107	88198	41.85957		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	19.45627	0.46494	73303	1751.17	<.0001
YesFamHistory	9.53683	0.36641	28358	677.45	<.0001
Vege	2.98757	0.23768	6613.51098	157.99	<.0001
Tech	-3.08933	0.24572	6616.68736	158.07	<.0001

Bounds on condition number: 1.0328, 9.1965

Figure 59: Stepwise Regression Step 3

Stepwise Selection: Step 4

Variable Dummy_BetwFood Entered: R-Square = 0.4016 and C(p) = 402.2308

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	54381	13595	353.29	<.0001
Error	2106	81043	38.48176		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	24.02906	0.55783	71403	1855.51	<.0001
YesFamHistory	8.77212	0.35576	23396	607.98	<.0001
Vege	3.38861	0.22978	8368.82747	217.48	<.0001
Dummy_BetwFood	-4.06356	0.29800	7155.53781	185.95	<.0001
Tech	-2.56173	0.23875	4430.16769	115.12	<.0001

Bounds on condition number: 1.0689, 16.82

Figure 60: Stepwise Regression Step 4

Stepwise Selection: Step 5

Variable Phy Entered: R-Square = 0.4312 and C(p) = 280.0619

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	58401	11680	319.21	<.0001
Error	2105	77023	36.59030		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	25.18825	0.55508	75344	2059.13	<.0001
YesFamHistory	8.38196	0.34890	21118	577.15	<.0001
Vege	3.46457	0.22418	8739.08273	238.84	<.0001
Dummy_BetwFood	-3.89012	0.29105	6536.55737	178.64	<.0001
Phy	-1.68818	0.16106	4019.99229	109.86	<.0001
Tech	-2.27284	0.23444	3439.13402	93.99	<.0001

Bounds on condition number: 1.0755, 26.369

Figure 61: Stepwise Regression Step 5

Stepwise Selection: Step 6

Variable Dummy_Alcohol Entered: R-Square = 0.4464 and C(p) = 218.7276

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	60451	10075	282.75	<.0001
Error	2104	74972	35.63314		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	23.57533	0.58759	57361	1609.77	<.0001
YesFamHistory	8.54329	0.34496	21856	613.35	<.0001
Vege	3.32691	0.22197	8004.55926	224.64	<.0001
Dummy_BetwFood	-3.78782	0.28754	6183.65381	173.54	<.0001
Phy	-1.59056	0.15946	3545.29436	99.49	<.0001
Tech	-2.15916	0.23184	3090.72040	86.74	<.0001
Dummy_Alcohol	1.93634	0.25526	2050.45983	57.54	<.0001

Bounds on condition number: 1.08, 37.943

Figure 62: Stepwise Regression Step 6

Stepwise Selection: Step 7

Variable YesHighCalories Entered: R-Square = 0.4573 and C(p) = 175.2375

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	61924	8846.25000	253.11	<.0001
Error	2103	73499	34.94978		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	21.18535	0.68862	33080	946.49	<.0001
YesFamHistory	8.13646	0.34734	19178	548.74	<.0001
YesHighCalories	2.72351	0.41955	1472.75139	42.14	<.0001
Vege	3.43190	0.22043	8471.87867	242.40	<.0001
Dummy_BetwFood	-3.61020	0.28608	5565.93603	159.26	<.0001
Phy	-1.47148	0.15899	2993.92437	85.66	<.0001
Tech	-2.14442	0.22961	3048.36609	87.22	<.0001
Dummy_Alcohol	1.78881	0.25382	1735.88860	49.67	<.0001

Bounds on condition number: 1.0907, 52.416

Figure 63: Stepwise Regression Step 7

Stepwise Selection: Step 8

Variable PublicTrans Entered: R-Square = 0.4647 and C(p) = 146.0096

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	62935	7866.84475	228.12	<.0001
Error	2102	72488	34.48543		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	20.01085	0.71760	26817	777.62	<.0001
YesFamHistory	8.27111	0.34592	19716	571.72	<.0001
YesHighCalories	2.64433	0.41701	1386.65211	40.21	<.0001
Vege	3.33223	0.21973	7930.86609	229.98	<.0001
Dummy_BetwFood	-3.57627	0.28424	5459.15066	158.30	<.0001
Phy	-1.39373	0.15858	2663.84937	77.25	<.0001
Tech	-2.24705	0.22887	3324.19121	96.39	<.0001
Dummy_Alcohol	1.77572	0.25214	1710.41199	49.60	<.0001
PublicTrans	1.61426	0.29814	1011.00803	29.32	<.0001

Bounds on condition number: 1.0923, 68.349

Figure 64: Stepwise Regression Step 8

Stepwise Selection: Step 9

Variable Age Entered: R-Square = 0.4893 and C(p) = 45.1349

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	66265	7362.81632	223.68	<.0001
Error	2101	69158	32.91660		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	12.30288	1.03861	4618.77528	140.32	<.0001
Age	0.25672	0.02552	3330.58885	101.18	<.0001
YesFamHistory	7.71651	0.34243	16716	507.82	<.0001
YesHighCalories	2.58702	0.40746	1326.93632	40.31	<.0001
Vege	3.33992	0.21468	7967.43134	242.05	<.0001
Dummy_BetwFood	-3.51802	0.27776	5280.47801	160.42	<.0001
Phy	-1.06464	0.15834	1488.05062	45.21	<.0001
Tech	-1.77908	0.22839	1997.31985	60.68	<.0001
Dummy_Alcohol	1.67646	0.24654	1522.09570	46.24	<.0001
PublicTrans	3.68142	0.35647	3510.68787	106.65	<.0001

Bounds on condition number: 1.6812, 97.748

Figure 65: Stepwise Regression Step 9

Stepwise Selection: Step 10

Variable MainMeal Entered: R-Square = 0.4947 and C(p) = 24.8369

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	10	66987	6698.72481	205.55	<.0001
Error	2100	68436	32.58851		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	11.43490	1.04974	3866.90064	118.66	<.0001
Age	0.26831	0.02551	3604.30654	110.60	<.0001
YesFamHistory	7.54327	0.34270	15789	484.50	<.0001
YesHighCalories	2.58718	0.40542	1327.10067	40.72	<.0001
Vege	3.21877	0.21515	7293.99617	223.82	<.0001
MainMeal	0.73354	0.15585	721.90117	22.15	<.0001
Dummy_BetwFood	-3.66395	0.27811	5656.45436	173.57	<.0001
Phy	-1.14437	0.15846	1699.61679	52.15	<.0001
Tech	-1.79287	0.22727	2028.05198	62.23	<.0001
Dummy_Alcohol	1.51879	0.24758	1226.38856	37.63	<.0001
PublicTrans	3.81861	0.35589	3751.86810	115.13	<.0001

Bounds on condition number: 1.6971, 120.46

Figure 66: Stepwise Regression Step 10

Stepwise Selection: Step 11

Variable YesMontCalories Entered: R-Square = 0.4977 and C(p) = 14.2862

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	67394	6126.68920	189.03	<.0001
Error	2099	68030	32.41046		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	11.94488	1.05673	4141.12130	127.77	<.0001
Age	0.26074	0.02553	3379.85320	104.28	<.0001
YesFamHistory	7.39875	0.34419	14976	462.09	<.0001
YesHighCalories	2.38487	0.40833	1105.59107	34.11	<.0001
Vege	3.26250	0.21492	7468.78557	230.44	<.0001
MainMeal	0.71534	0.15551	685.77573	21.16	<.0001
Dummy_BetwFood	-3.60971	0.27777	5473.50453	168.88	<.0001
YesMontCalories	-2.18920	0.61828	406.33311	12.54	0.0004
Phy	-1.12514	0.15812	1641.04081	50.63	<.0001
Tech	-1.80167	0.22666	2047.77837	63.18	<.0001
Dummy_Alcohol	1.53753	0.24696	1256.27048	38.76	<.0001
PublicTrans	3.76381	0.35525	3638.03753	112.25	<.0001

Figure 67: Stepwise Regression Step 11

Stepwise Selection: Step 12

Variable ActiveTrans Entered: R-Square = 0.4987 and C(p) = 11.7164

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	12	67542	5628.46060	173.96	<.0001
Error	2098	67882	32.35539		
Corrected Total	2110	135423			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	11.37518	1.08893	3530.71604	109.12	<.0001
Age	0.27298	0.02615	3527.05940	109.01	<.0001
YesFamHistory	7.39239	0.34391	14950	462.04	<.0001
YesHighCalories	2.50848	0.41206	1199.09484	37.06	<.0001
Vege	3.21186	0.21603	7151.75400	221.04	<.0001
MainMeal	0.72543	0.15545	704.59568	21.78	<.0001
Dummy_BetwFood	-3.61761	0.27756	5496.51864	169.88	<.0001
YesMontCalories	-2.16256	0.61788	396.34374	12.25	0.0005
Phy	-1.14472	0.15825	1692.96879	52.32	<.0001
Tech	-1.86346	0.22830	2155.54674	66.62	<.0001
Dummy_Alcohol	1.52115	0.24687	1228.45224	37.97	<.0001
ActiveTrans	1.83983	0.86040	147.94603	4.57	0.0326
PublicTrans	4.06182	0.38133	3671.02525	113.46	<.0001

Bounds on condition number: 1.7951, 177.86

Figure 68: Stepwise Regression Step 12

All variables left in the model are significant at the 0.0500 level.

No other variable met the 0.1500 significance level for entry into the model.

Summary of Stepwise Selection

Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	YesFamHistory		1	0.2338	0.2338	1098.03	643.48	<.0001
2	Tech		2	0.0661	0.2999	823.527	199.03	<.0001
3	Vege		3	0.0488	0.3487	621.250	157.99	<.0001
4	Dummy_BetwFood		4	0.0528	0.4016	402.231	185.95	<.0001
5	Phy		5	0.0297	0.4312	280.062	109.86	<.0001
6	Dummy_Alcohol		6	0.0151	0.4464	218.728	57.54	<.0001
7	YesHighCalories		7	0.0109	0.4573	175.237	42.14	<.0001
8	PublicTrans		8	0.0075	0.4647	146.010	29.32	<.0001
9	Age		9	0.0246	0.4893	45.1349	101.18	<.0001
10	MainMeal		10	0.0053	0.4947	24.8369	22.15	<.0001
11	YesMontCalories		11	0.0030	0.4977	14.2862	12.54	0.0004
12	ActiveTrans		12	0.0011	0.4987	11.7164	4.57	0.0326

Figure 69: Summary of Stepwise Regression Step

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: BMI

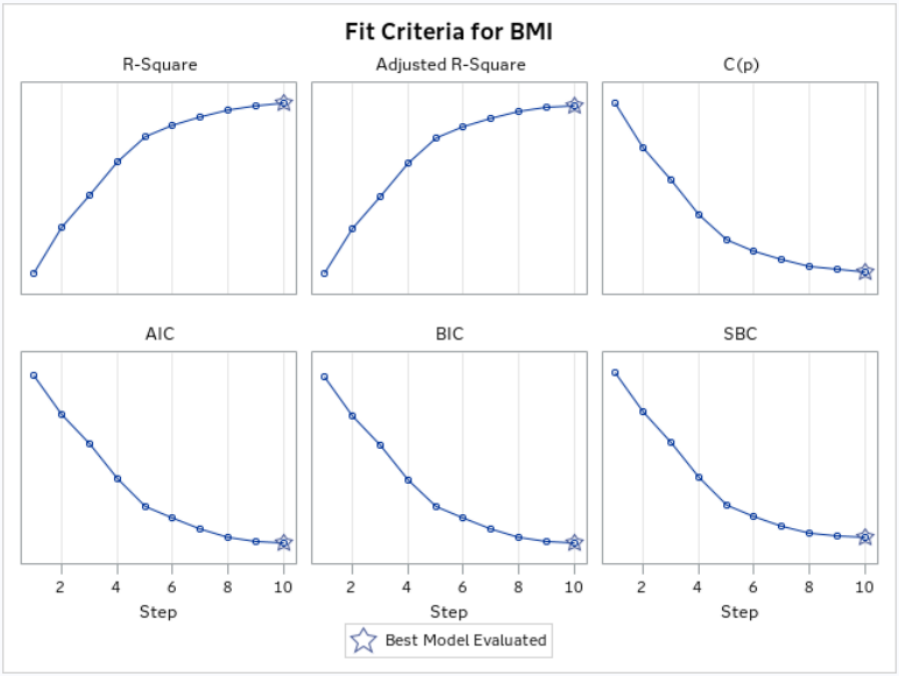


Figure 70: Fit Criteria for BMI

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation	95% Confidence Limits	
Intercept	1	11.37518	1.08893	10.45	<.0001	.	0	9.23968	13.51067
Age	1	0.27298	0.02615	10.44	<.0001	0.55707	1.79509	0.22170	0.32425
YesFamHistory	1	7.39239	0.34391	21.50	<.0001	0.86905	1.15068	6.71795	8.06683
YesHighCalories	1	2.50848	0.41206	6.09	<.0001	0.87992	1.13647	1.70040	3.31656
Vege	1	3.21186	0.21603	14.87	<.0001	0.91451	1.09349	2.78820	3.63553
MainMeal	1	0.72543	0.15545	4.67	<.0001	0.92048	1.08639	0.42057	1.03028
Dummy_BetwFood	1	-3.61761	0.27756	-13.03	<.0001	0.90670	1.10291	-4.16193	-3.07330
YesMontCalories	1	-2.16256	0.61788	-3.50	0.0005	0.92486	1.08124	-3.37429	-0.95084
Phy	1	-1.14472	0.15825	-7.23	<.0001	0.88282	1.13273	-1.45507	-0.83438
Tech	1	-1.86346	0.22830	-8.16	<.0001	0.86694	1.15348	-2.31119	-1.41573
Dummy_Alcohol	1	1.52115	0.24687	6.16	<.0001	0.94684	1.05614	1.03702	2.00528
ActiveTrans	1	1.83983	0.86040	2.14	0.0326	0.80174	1.24728	0.15251	3.52716
PublicTrans	1	4.06182	0.38133	10.65	<.0001	0.55987	1.78614	3.31399	4.80964

Collinearity Diagnostics

			Proportion of Variation											
Number	Eigenvalue	Condition Index	Intercept	Age	YesFamHistory	YesHighCalories	Vege	MainMeal	Dummy_BetwFood	YesMontCalories	Phy	Tech	Dummy_Alcohol	ActiveTrans	PublicTrans
1	8.49175	1.00000	0.00017499	0.00044473	0.00181	0.00124	0.00207	0.00239	0.00158	0.00067864	0.00337	0.00291	0.00308	0.00036671	0.00148
2	1.12299	2.74986	0.00001885	0.00019718	0.00229	0.00170	0.00008456	0.00004994	0.00009063	0.14229	0.02724	0.04304	0.00216	0.41671	0.00326
3	0.93794	3.00892	5.096833E-7	0.00003114	0.00106	0.00019439	0.00010170	0.00013249	0.00003712	0.72895	0.00010233	0.01074	0.00004980	0.15824	0.00122
4	0.67928	3.53568	0.00002626	0.00080998	0.00217	0.00021251	0.00826	0.00087674	0.00009051	0.00670	0.01450	0.65552	0.01653	0.14747	0.00310
5	0.51106	4.07627	0.00002073	0.00015311	0.00192	0.00133	0.00004197	0.00137	0.00007057	0.02510	0.79192	0.09536	0.01841	0.05335	0.00145
6	0.29728	5.34464	0.00037189	0.00183	0.05593	0.00332	0.00999	0.00062635	0.00758	0.00605	0.02548	0.01313	0.84200	0.00492	0.00212
7	0.25371	5.78534	0.00001920	0.01533	0.05604	0.00631	0.03752	0.01680	0.00097355	0.02272	0.00004028	0.03507	0.00314	0.06923	0.36819
8	0.20791	6.39091	0.00045069	0.00242	0.08133	0.03758	0.05247	0.62155	0.02844	0.01274	0.06272	0.00028552	0.02556	0.01947	0.01670
9	0.17222	7.02197	0.00016676	0.00844	0.11332	0.00530	0.29776	0.30834	0.13454	0.02837	0.00055574	0.01189	0.00049733	0.04349	0.07427
10	0.14630	7.61859	0.00119	0.00346	0.10173	0.01686	0.52548	6.448002E-7	0.34908	0.00205	0.00600	0.09679	0.00367	0.01042	0.00596
11	0.10485	8.99950	0.00071164	0.00220	0.43738	0.50328	0.01551	0.01605	0.17449	0.00318	0.00020043	0.00075982	0.06132	0.00596	0.00400
12	0.06479	11.44857	0.01833	0.25797	0.14030	0.32654	0.04145	0.00298	0.24044	0.00382	0.00251	0.00318	0.02081	0.00234	0.07090
13	0.00993	29.24475	0.97852	0.70671	0.00473	0.09614	0.00926	0.02883	0.06258	0.01734	0.06535	0.03133	0.00276	0.06803	0.44735

Figure 71: Parameter Estimates and Collinearity Table

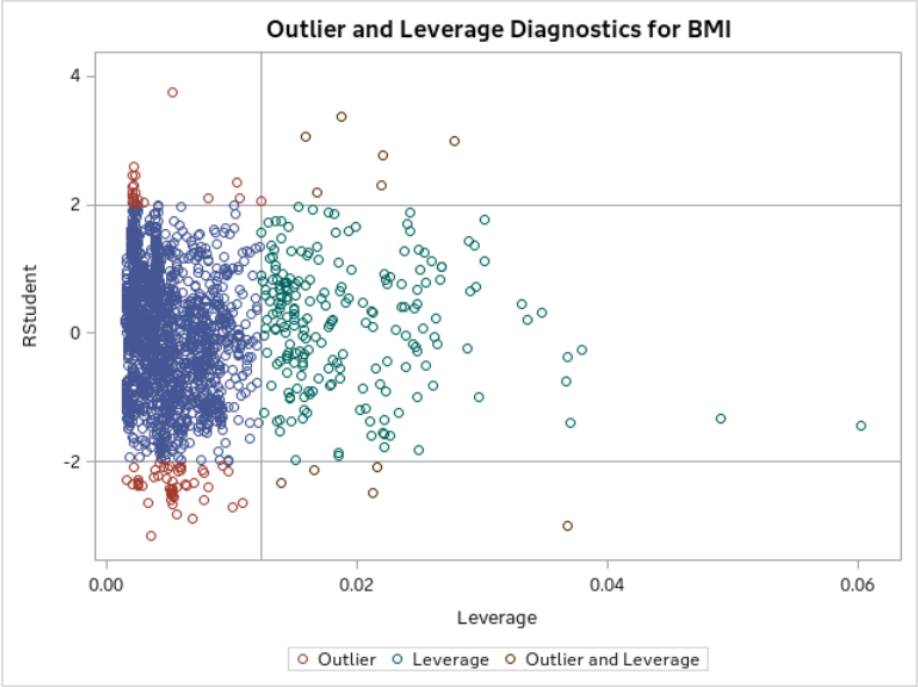


Figure 72: Outlier and Leverage Diagnostics for BMI

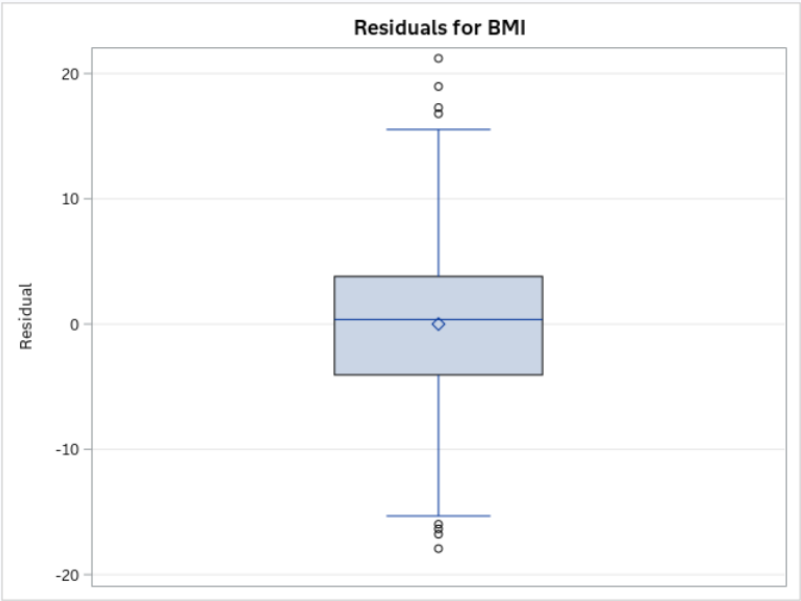


Figure 73: Residuals of BMI

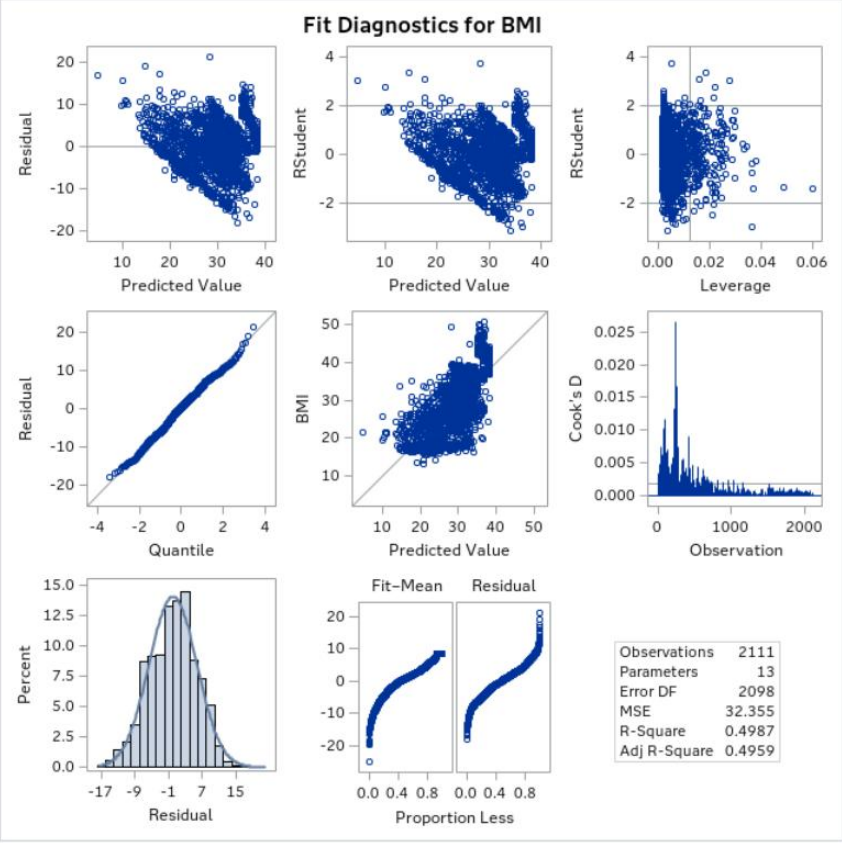


Figure 74: Fit Diagnostics for BMI

Linear Regression Results

The REG Procedure
Model: Linear_Regression_Model
Dependent Variable: BMI

Number of Observations Read	2111
Number of Observations Used	2111

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	12	67542	5628.46060	173.96	<.0001
Error	2098	67882	32.35539		
Corrected Total	2110	135423			

Root MSE	5.68818	R-Square	0.4987
Dependent Mean	29.70015	Adj R-Sq	0.4959
Coeff Var	19.15202		

Parameter Estimates									
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation	95% Confidence Limits	
Intercept	1	11.37518	1.08893	10.45	<.0001	.	0	9.23968	13.51067
Age	1	0.27298	0.02615	10.44	<.0001	0.55707	1.79509	0.22170	0.32425
YesFamHistory	1	7.39239	0.34391	21.50	<.0001	0.86905	1.15068	6.71795	8.06683
YesHighCalories	1	2.50848	0.41206	6.09	<.0001	0.87992	1.13647	1.70040	3.31656
Vege	1	3.21186	0.21603	14.87	<.0001	0.91451	1.09349	2.78820	3.63553
MainMeal	1	0.72543	0.15545	4.67	<.0001	0.92048	1.08639	0.42057	1.03028
Dummy_BetwFood	1	-3.61761	0.27756	-13.03	<.0001	0.90670	1.10291	-4.16193	-3.07330
YesMontCalories	1	-2.16256	0.61788	-3.50	0.0005	0.92486	1.08124	-3.37429	-0.95084
Phy	1	-1.14472	0.15825	-7.23	<.0001	0.88282	1.13273	-1.45507	-0.83438
Tech	1	-1.86346	0.22830	-8.16	<.0001	0.86694	1.15348	-2.31119	-1.41573
Dummy_Alcohol	1	1.52115	0.24687	6.16	<.0001	0.94684	1.05614	1.03702	2.00528
ActiveTrans	1	1.83983	0.86040	2.14	0.0326	0.80174	1.24728	0.15251	3.52716
PublicTrans	1	4.06182	0.38133	10.65	<.0001	0.55987	1.78614	3.31399	4.80964

Figure 75: Multiple Linear Regression Result

The model was statistically significant with $F(12, 2098) = 173.96$ and $p < 0.0001$, explaining 49.87% of the variation in BMI ($R^2 = 0.4987$). In performance, it closely mirrored Model 3. Both procedures retained the same predictor set and yielded nearly identical coefficient estimates. To be exact, YesFamHistory was the strongest positive predictors with $\beta = 7.39$ and $p < 0.0001$. On the other hand, Dummy_BetwFood was the strongest negative predictor with $\beta = -3.62$ and $p < 0.0001$. Besides, ActiveTrans and PublicTrans transportation variables were found to have significant effect.

The results of this validation analysis indicate that the findings from Model 3 are stable and robust. The fact that the manual refinement and the automated stepwise process yielded matching predictors shows that the selection for Model 3 was solid rather than dependent on a specific technique.

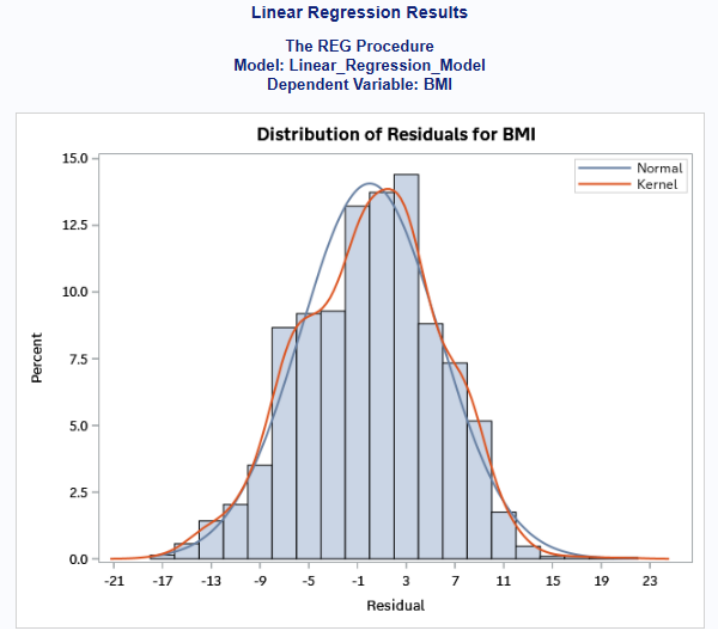


Figure 76: Distribution of Residuals for BMI Linear Regression Model

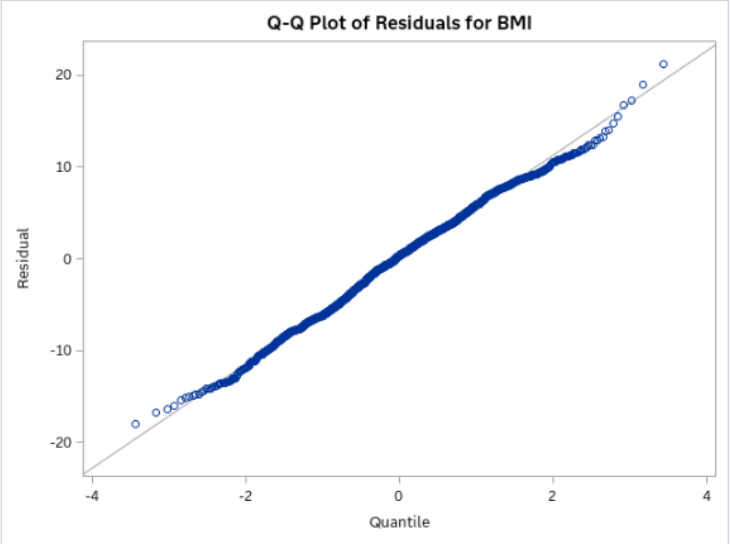


Figure 77: Q-Q Plot of Residuals for BMI

The residual histogram (Figure 76) illustrates that the data approximates a normal distribution, with most of the residuals being located near the mean while outliers were sparse. This can be seen in the Q-Q plot (Figure 77) as well, most points fell along the line although some of the extreme values deviated in the tails. Due to the large sample size (n = 2111), these deviations were not severe enough to impact the regression's results.

All in all, the stepwise results back up the stability of Model 3 and gives us strong confidence in the final regression findings.

References

UCI Machine Learning Repository (2019). *Estimation of obesity levels based on eating habits and physical condition* [Data set]. https://doi.org/10.24432/C5H31Z

Hildebrand, S., & Pfeifer, A. (2025). The obesity pandemic and its impact on non-communicable disease burden. *Pflugers Arch - Eur J Physiol*, *477*, 657–668. https://doi.org/10.1007/s00424-025-03066-8

Palechor, F. M., & de la Hoz Manotas, A. (2019). Dataset for estimation of obesity levels based on eating habits and physical condition in individuals from Colombia, Peru and Mexico. *Data in Brief*, *25*, Article 104344. https://doi.org/10.1016/j.dib.2019.104344

World Health Organization. (2024, February 23). *The challenge of obesity*. Www.Who.Int. https://www.who.int/europe/news-room/fact-sheets/item/the-challenge-of-obesity